Magic Gems: Convex Hulls of Lifted Magic Squares

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Abstract

To an arrangement of $1, \dots, n^2$ in an $n \times n$ grid we attach the n^2 points (x, y, z) in $\mathbb{R}^3$, where (x, y) is the position of a cell measured from the centre of the grid and z is its entry minus the mean entry. Their convex hull is the *gem* of the arrangement; its lower and upper faces are the regular subdivisions of the grid induced by the heights and by their negatives. Magic squares are the motivating case. Our main result concerns the Siamese (de la Loubère) square of every odd order $n \geq 5$: its gem has exactly the twelve vertices that the boundary of the grid forces (the corners and, on each side, the cells of the smallest and largest entry), with f-vector $(12, 26, 16)$ and isometry group $\{\pm\mathrm{id}\}$. The proof shears the lift onto five levels and reduces to the signs of a table of polynomials in $m = (n-1)/2$. The same method gives the f-vector $(12, 24, 14)$ for the square with the two base-n digits exchanged, and we determine and count the magic linear squares. Along every family of linear squares with fixed steps and offsets affine in n the number of vertices is bounded and the f-vector is eventually periodic in n, while over all linear magic squares of prime order n the largest number of vertices is of order $\Theta(\ln n)$.

Most other results need less than magic. If the heights have vanishing first moments, as for every semi-magic square, the point set has a non-trivial isometry exactly when the arrangement equals its complement up to rotation and reflection. For the 880 magic squares of order four a computational census gives 616 congruence classes and 352 combinatorial types of gems, none with an isometry that its point set lacks. The gem hardly sees the magic conditions: at order four the mean numbers of vertices and of lower and upper facets of magic gems differ from those of semi-magic squares and of random arrangements by at most 0.14 standard deviations, and the excess of vertices over semi-magic squares comes almost entirely from self-complementary squares; at order five, on samples, the mean vertex counts differ by less than 0.07 vertices. The volume and the solid centroid do shift.

Keywords: magic squares, convex polytopes, regular subdivisions, symmetry groups, Siamese method, linear magic squares, integer hulls
MSC 2020: 05B15, 52B05, 52B15, 52B20

1 Introduction

Magic squares are among the oldest objects of recreational mathematics: the 3×3 Lo Shu belongs to Chinese tradition, a 4×4 square is engraved in Dürer's *Melencolia I* (1514), and constructions for every order $n \geq 3$ have been known for centuries [2, 28]. The modern literature treats them from several directions: enumeration and the geometry of the polytopes whose lattice points are the squares [35, 14, 1, 7, 6], spectral and matrix properties [22, 24, 25, 9], and, in work of Loly and collaborators, physical invariants such as the moment of inertia and the multipole moments [21, 30, 31]. Magic squares have also been read as height fields: Knecht's water-retention problem asks how much water a magic square of order n can hold when its entries are the heights of the cells [26], and led to the study of water retention on random surfaces [19].

Definition 1.1 (Magic square)**.** An *arrangement* of order n is a bijection $S = (s_{ij})$ from the cells of the $n \times n$ grid onto $\{1, \dots, n^2\}$ (a permutation of $1, \dots, n^2$ written into the grid). It is a *magic square* if every row, every column and both main diagonals sum to the magic constant $\mu_n = n(n^2+1)/2$, and *semi-magic* if only the rows and columns do.

The construction. Centre the grid at the origin and attach to the cell (i, j) of an arrangement the point

$$\mathbf{p}_{ij} = (x_{ij}, y_{ij}, z_{ij}),$$

whose horizontal part (x_{ij}, y_{ij}) is the position of the cell and whose height $z_{ij} = s_{ij} - \frac{n^2+1}{2}$ is the entry minus the mean entry. The *gem* $\mathcal{G}(S)$ is the convex hull of these n^2 points (Section 2). It is defined for every arrangement: the points are the lift of a fixed grid by a permutation of fixed heights, and the lower and upper faces of the gem are the regular subdivisions $\mathcal{S}(z)$ and $\mathcal{S}(-z)$ of the grid, glued along its boundary (Proposition 5.2). Magic squares are the motivating case, and we call the gem of a magic square a *Magic Gem*: read as vectors from the origin, its points have vertical components that cancel along every row, column and both diagonals (Lemma 2.2; Figures 1 and 2). Figure 1 shows Dürer's square, its vectors and its gem, which has 12 vertices and 4 points in its interior.

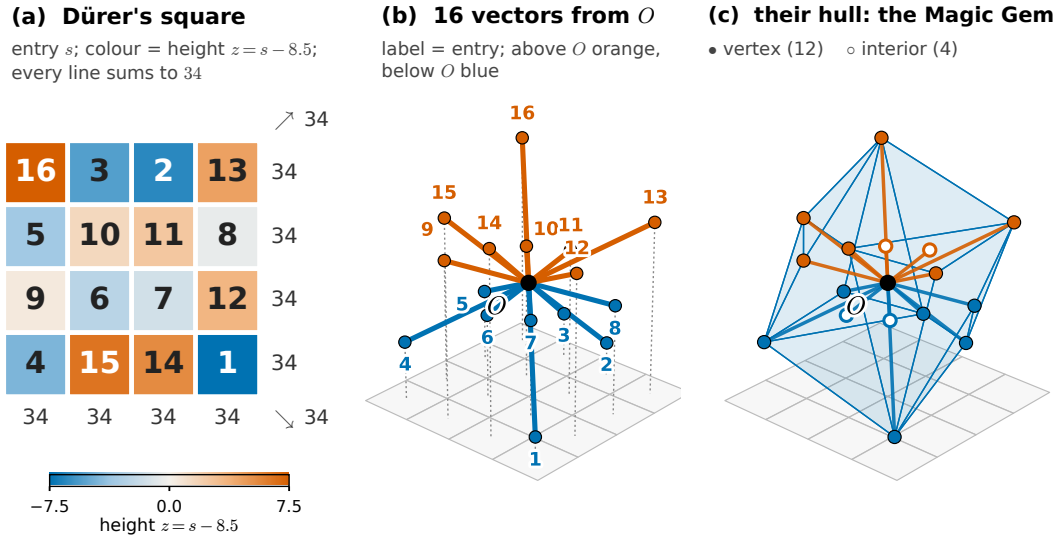


Figure 1: Dürer's square. (a) Entries s_{ij} on a heat map of the heights $z_{ij} = s_{ij} - 8.5$ (orange above 0, blue below); every line sums to 34. (b) The 16 vectors $\mathbf{p}_{ij}$ from the origin O, labelled by the entries, with drop lines to the floor grid. (c) Their convex hull, the Magic Gem: 12 tips are vertices (filled), 4 lie in the interior (open). The colour bar is shared by (a) and (b).

Results. The main result is a theorem about one classical family. *For the Siamese (de la Loubère) square of every odd order $n \geq 5$ the vertices of the gem are exactly the twelve points that the boundary of the grid forces (the corners and, on each side, the cells of the smallest and the largest entry); the f-vector is* $(12, 26, 16)$ *and the isometry group is* $\{\pm\mathrm{id}\}$ (Theorem 6.4). So only 12 of the n^2 points are vertices, whereas the gems of random arrangements and of sampled magic squares have more and more vertices as n grows (Table 7, estimates). The proof shears the lift onto five levels and reduces to the signs of a table of polynomials in $m = (n-1)/2$. The same method shows that the square with the two base-n digits exchanged has a gem with f-vector $(12, 24, 14)$ for every odd $n \geq 5$ (Proposition 6.5); along the way we give a self-contained criterion for a linear square (a square of the uniform step method) to be magic, and count these squares (Proposition 6.1). More generally, along every family of linear squares with fixed steps and offsets affine in n the vertex count is bounded and the f-vector is eventually periodic in n (Proposition 6.7). The bound depends on the family: over all linear magic squares of prime order n the largest vertex count grows like $\Theta(\ln n)$ and is at least $\lceil \ln n/(4 \ln \varphi) \rceil + 2$, with φ the golden ratio (Corollary 6.10), and, by computation, for $7 \leq n \leq 21$ the Siamese and the digit-exchanged squares are the only linear magic squares, up to D_4, with as few as twelve vertices (Remark 6.6).

The other results need weaker hypotheses than magic, and most of them hold for every arrangement; Table 1 lists what each one needs. The points balance about the origin for every arrangement, and the heights balance as weights on the grid exactly when the rows and columns

have equal sums (Section 3). If only the first moments $\sum z\,x$ and $\sum z\,y$ of the heights vanish, the covariance matrix of the point set is $\operatorname{diag}(a_n, a_n, b_n)$ with $a_n \neq b_n$, and a short principal-axes argument shows that the point set has a non-trivial isometry exactly when the arrangement equals its complement up to D_4 (Proposition 4.2). For the 880 magic squares of order four up to rotation and reflection we compute all gems exactly, by computer: the isometry group of each gem equals that of its point set, so 48 gems are centrally symmetric, 304 have one half-turn and 528 have none (Proposition 4.3); the gems have 8 to 16 vertices, 39 f-vectors, 352 combinatorial types and 155 distinct volumes, and form 616 congruence classes (Proposition 5.4); and the vertex set determines the square, for 871 of the 880 classes already by the rank of the line equations (Remark 5.5).

Finally, Section 5.3 compares magic gems with the gems of semi-magic squares, of arrangements with vanishing first moments and of random arrangements: the mean numbers of vertices and facets hardly differ, while the volume and the position of the solid centroid change with the line conditions.

Table 1: The results and the hypotheses they need. *Except the eight D_4-images of the natural square $s_{ij} = ni + j + 1$, whose points are coplanar.

result	statement	hypothesis
Lemma 2.2, Lemma 5.1, Lemma 4.1	centring; forced vertices; complements	any arrangement
Proposition 2.3, Proposition 5.2	3-polytope; faces $\mathcal{S}(z)$, $\mathcal{S}(-z)$	any arrangement*
Lemma 3.1, Proposition 4.2	covariance; isometries of point sets	vanishing first moments
Lemma 4.4	parity of V, V^∂, V°; of E, F	self-complementary*; associative*
Proposition 4.3, 5.4, Remark 5.5	order-four census	magic, $n = 4$ (computation)
Section 5.3	which invariants see the line conditions	$n = 4, 5$ (computation)
Proposition 6.1	magic linear squares	linear squares, odd n
Theorem 6.4, Proposition 6.5	Siamese and digit-exchanged gems	two magic squares, odd $n \geq 5$
Proposition 6.7, Corollary 6.10	fixed steps; largest vertex count	linear squares; prime n

Relation to previous work. The polyhedral geometry of magic squares is classical: nonnegative integer matrices with all line sums equal to t are the lattice points of the t-th dilate of the Birkhoff polytope, and their number is a polynomial in t [35, 14, 38]; the count of magic squares with distinct entries is an Ehrhart quasi-polynomial of an inside-out polytope [7, 6]; the cones of magic squares and cubes have Hilbert-basis descriptions [1]; contingency tables with fixed margins are the same lattice points seen as a sample space [12]. In all of these the magic squares are the *points* of a polytope, whereas the gem attaches a polytope to a *single* arrangement. The lifting construction itself is that of regular subdivisions and secondary fans [17, 11]. The moment identities of Section 3 are due in substance to Loly and Rogers [21, 30, 31]; we record them in the coordinates of the gem for later use. The Siamese square is de la Loubère's construction [10], and the linear squares are the squares of the uniform step method [2, Ch. I], analysed by Lehmer [20] and by Apostol and Zuckerman [3]; we give a self-contained proof of Proposition 6.1 but do not claim priority for it. For linear squares with fixed steps we use integer hulls of linear families of polygons, a planar case of a theorem of Calegari and Walker [8], and the lower bound rests on Rubin's Fibonacci triangle [33]. The partition of the 880 order-four squares into associative, self-complementary and other squares is Dudeney's [13]. New are the Siamese theorem and its companion for the digit-exchanged square, the results on linear squares with fixed steps and on their largest vertex count, the exact order-four census and the comparison of Section 5.3, together with a short covariance argument that settles symmetry and congruence of the point sets for every order.

Sections 2–7 treat, in turn, the gem, the balance of its vectors, symmetry and congruence, faces and the order-four census, the Siamese and linear squares, and open questions; the appendices describe the computations and collect lattice-point counts, vertex statistics and further order-four data. An interactive display of the 880 order-four gems is available at `https://kylemath.github.io/MagicGemWebpage/`.

2 The gem of an arrangement

Let $S = (s_{ij})$ be an arrangement of order $n \geq 2$, with cells indexed by $i, j \in \{0, \ldots, n-1\}$ (i the row from the top, j the column from the left), and write $N = n^2$ and $\bar{s} = (N+1)/2$ for the mean entry.

Definition 2.1 (Lifted point set and gem)**.** To the cell (i, j) assign the point $\mathbf{p}_{ij} = (x_{ij}, y_{ij}, z_{ij}) \in \mathbb{R}^3$ with

$$x_{ij} = j - \frac{n-1}{2}, \qquad y_{ij} = \frac{n-1}{2} - i, \qquad z_{ij} = s_{ij} - \bar{s}. \tag{1}$$

The *gem* of S is the convex hull $\mathcal{G}(S) = \operatorname{conv}\{\mathbf{p}_{ij}\}$, and $z = (z_{ij})$ is the *height function* of S; for a magic square we also call it the *Magic Gem*.

The x-axis runs along the columns and the y-axis along the rows (top row at positive y). Let $x_j = j - \frac{n-1}{2}$ and $y_i = \frac{n-1}{2} - i$ denote the n values taken by x and y. A *line* is one of the n rows, the n columns, the main diagonal $\{i = j\}$ or the anti-diagonal $\{i + j = n-1\}$. For a line L let $\delta_L(S) = \sum_{c \in L} s_c - \mu_n$ be the *excess* of the line sum over the magic constant, so S is magic iff $\delta_L(S) = 0$ for all $2n+2$ lines. Moment and centroid statements below refer to the N points $\mathbf{p}_{ij}$ with unit weights, not to the solid $\mathcal{G}(S)$.

Terminology. Each point $\mathbf{p}_{ij}$ is either a *vertex* of $\mathcal{G}(S)$, an *interior point* (in the interior of $\mathcal{G}(S)$), or a *non-vertex boundary point* (on the boundary but not a vertex). We write $\operatorname{vert}\mathcal{G}(S)$ for the vertex set, V, V°, V^∂ for the three numbers ($V + V^\circ + V^\partial = N$), and (V, E, F) for the f-vector. For any set $X \subseteq \mathbb{R}^3$, $\operatorname{Iso}(X)$ is the group of isometries of $\mathbb{R}^3$ that map X onto itself. The dihedral group $D_4 = \langle r, h \rangle$ of the square acts on arrangements by $(gS)_{gc} = s_c$; here r is the quarter-turn, r^2 the half-turn, h and v the left–right and top–bottom reflections and d the reflection in the main diagonal $i = j$. For $g \in D_4$ let $\hat{g}(x, y, z) = (g(x, y), z)$, where g acts on the plane as the corresponding orthogonal map. The order-four squares are numbered $\#0, \ldots, \#879$ in the lexicographic order of their Frénicle standard forms (the image under D_4 with the smallest corner entry at top left and $s_{01} < s_{10}$) [15]; the list is `anc/squares880.txt`.

Lemma 2.2 (Means, variances, line sums)**.** *For every arrangement S of order n:*

1. $\sum_{i,j} x_{ij} = \sum_{i,j} y_{ij} = \sum_{i,j} z_{ij} = 0$ *and* $\sum_{i,j} x_{ij} y_{ij} = 0$*;*
2. $\frac{1}{N}\sum x_{ij}^2 = \frac{1}{N}\sum y_{ij}^2 = a_n := \frac{n^2-1}{12}$ *and* $\frac{1}{N}\sum z_{ij}^2 = b_n := \frac{N^2-1}{12}$*;*
3. *for every line L, $\delta_L(S) = \sum_{c \in L} z_c$, and $\sum_{c \in L} \mathbf{p}_c = \big(n\,\bar{x}_L,\ n\,\bar{y}_L,\ \delta_L(S)\big)$, where $(\bar{x}_L, \bar{y}_L)$ is the centre of the line: $(0, y_i)$ for the row i, $(x_j, 0)$ for the column j, and $(0, 0)$ for both diagonals.*

In particular S is magic if and only if the vector sum of the $\mathbf{p}_c$ along every line is horizontal.

Proof. (1) $\sum_{i,j} x_{ij} = n\sum_j x_j = 0$, likewise for y; $\sum z = \sum_{k=1}^{N} k - N\bar{s} = 0$; and $\sum xy = (\sum_j x_j)(\sum_i y_i) = 0$. (2) $\frac{1}{n}\sum_j x_j^2$ is the variance of n equally spaced points with unit spacing, $(n^2-1)/12$; the same holds for y, and for z with N points. (3) Since $\mu_n = n\bar{s}$, $\sum_{c \in L} z_c = \delta_L$. The cells of row i have $y = y_i$ and x running over all x_j; for a column the roles are exchanged; each diagonal is invariant under $(x, y) \mapsto (-x, -y)$. □

For $g \in D_4$ the points of gS are $\hat{g}\,\mathbf{p}_c$, so

$$\mathcal{G}(gS) = \hat{g}\,\mathcal{G}(S), \tag{2}$$

and since D_4 maps magic squares to magic squares, every D_4-class of magic squares has a single gem up to congruence.

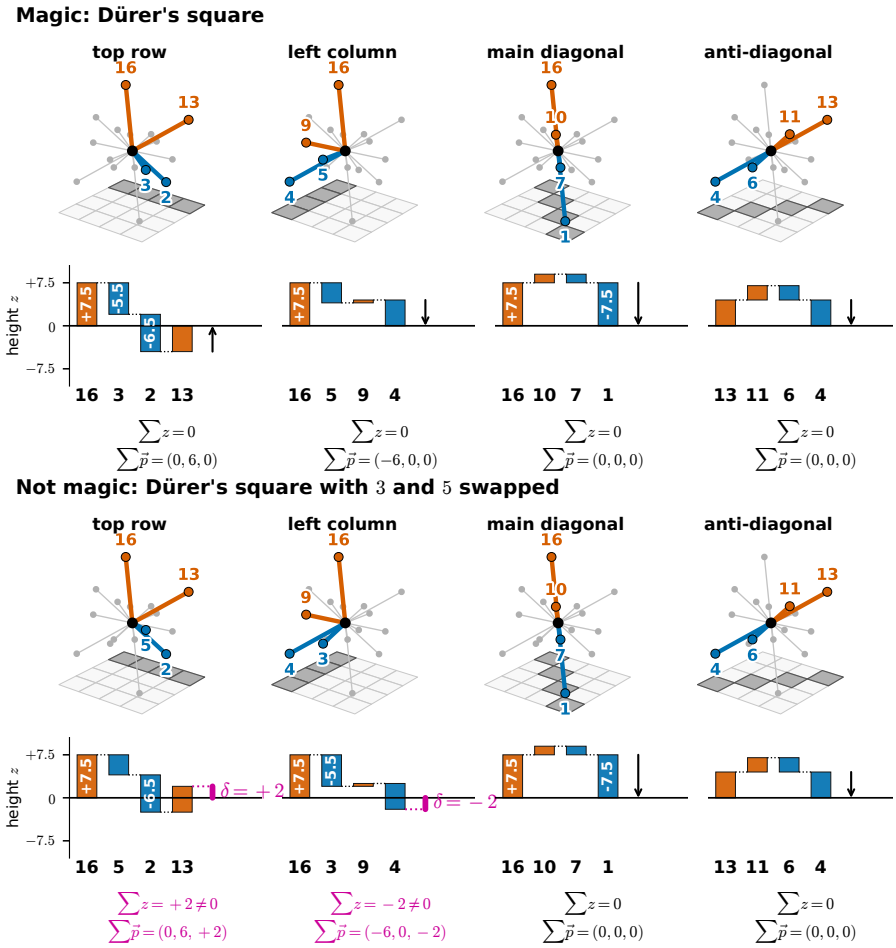


Figure 2: Heights cancel along every line. *Top:* for Dürer's square, the vectors of the top row, left column and both diagonals (coloured; the cells of the line shaded on the floor), and below each the heights $z = s - 8.5$ of the line stacked head to tail; the running total returns to 0, so $\sum_{c \in L} \mathbf{p}_c$ is horizontal (Lemma 2.2(3)). *Bottom:* after swapping the entries 3 and 5 the top row and the left column no longer close (excesses $\delta_L = \pm 2$, magenta); the diagonals do not contain the swapped cells.

Proposition 2.3 (Dimension)**.** *Let S be an arrangement of order $n \geq 2$. The points $\mathbf{p}_{ij}$ are coplanar if and only if S is one of the eight D_4-images of the natural square $s_{ij} = ni + j + 1$. Otherwise $\mathcal{G}(S)$ is a 3-polytope with the origin in its interior. In particular this holds for every semi-magic square.*

Proof. The grid positions are not collinear, so a plane containing all points is not vertical and the points are coplanar if and only if $z = \alpha x + \beta y + \gamma$ on the grid. Then $u = \alpha$ and $w = -\beta$ are the differences of the entries of horizontally and vertically adjacent cells, so they are integers, and $s_{ij} = s_{00} + uj + wi$. Since the entries are distinct, $u, w \neq 0$; applying h, v and d we may assume $1 \leq u \leq w$, so that $s_{00} = 1$ is the smallest entry. The entry 2 is $1 + uj + wi$ with $(i, j) \neq (0, 0)$, hence $u = 1$, and row 0 holds $1, \ldots, n$. The entry $n+1$ lies in a row $i \geq 1$, so $w \leq n$; if $w < n$ then the cells $(0, w)$ and $(1, 0)$ both hold $1 + w$. Hence $w = n$ and $s_{ij} = ni + j + 1$; conversely, every D_4-image of this arrangement has affine heights. If the points span $\mathbb{R}^3$ affinely, their centroid, which is $\mathbf{0}$ by Lemma 2.2(1), is a convex combination with positive weights of all of them and hence an interior point of the hull. Finally, if S is semi-magic and $z = \alpha x + \beta y + \gamma$, then $\gamma = 0$ by Lemma 2.2(1), the row i has height sum $n\beta y_i = 0$ for all i and the column j has $n\alpha x_j = 0$, so $\alpha = \beta = 0$, which is impossible. □

Example 2.4 (Lo Shu)**.** For the Lo Shu square $S = \begin{pmatrix} 2 & 7 & 6 \\ 9 & 5 & 1 \\ 4 & 3 & 8 \end{pmatrix}$, $\bar{s} = 5$ and the nine points are,

row by row, $(-1,1,-3)$, $(0,1,2)$, $(1,1,1)$, $(-1,0,4)$, $(0,0,0)$, $(1,0,-4)$, $(-1,-1,-1)$, $(0,-1,-2)$, $(1,-1,3)$. The origin is the only non-vertex point and is interior; the gem has f-vector $(8,16,10)$ and volume 18 (Figure 3).

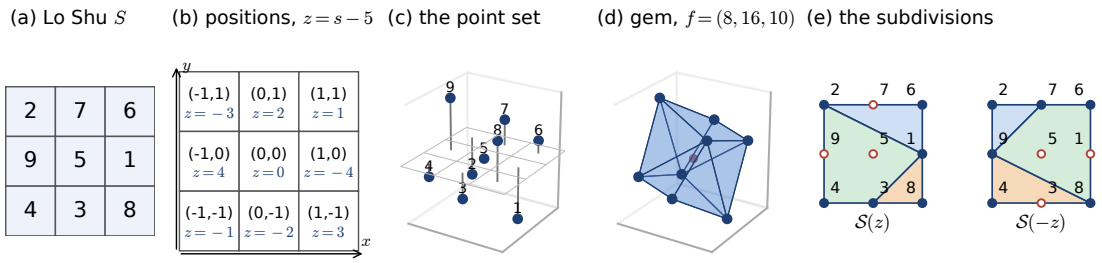


Figure 3: The gem of the Lo Shu square: (a) the square; (b) the cell positions (x_{ij}, y_{ij}), measured from the centre, with the heights $z_{ij} = s_{ij} - 5$; (c) the nine points, labelled by the entries; (d) their convex hull, with f-vector $(8,16,10)$ and the interior point $\mathbf{0}$; (e) the regular subdivisions $\mathcal{S}(z)$ and $\mathcal{S}(-z)$ of the grid, the projections of the lower and the upper facets (Example 5.3); grid points that belong to no cell are drawn as open circles.

3 Balance: the vectors from the centroid

Figures 1 and 2 draw the points $\mathbf{p}_{ij}$ as vectors from the origin. This section records which balance statements these vectors satisfy, and for which arrangements. The identities are due in substance to Loly and Rogers [21, 30, 31], who study the moment of inertia and the multipole moments of magic squares and cubes.

Unweighted balance holds for every arrangement. By Lemma 2.2(1), $\sum_{i,j} \mathbf{p}_{ij} = \mathbf{0}$ for every arrangement: the origin is the centroid of the N points, and by Proposition 2.3 it is an interior point of $\mathcal{G}(S)$ unless S is a D_4-image of the natural square. It is a consequence of centring and concerns the points, not the solid $\mathcal{G}(S)$, whose centre of mass need not be the origin (Appendix C).

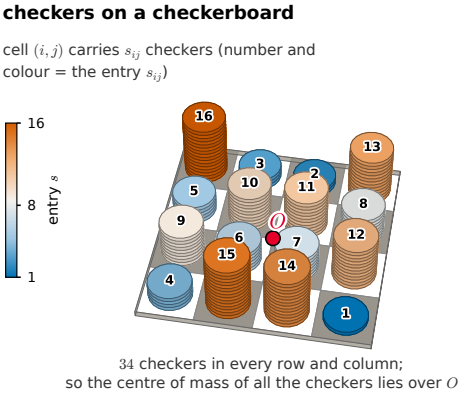


Figure 4: Dürer's square as piles of checkers, s_{ij} on cell (i,j), seen from the front of the board in the orientation of the square; each row and column carries 34 checkers, so the centre of mass of all the checkers lies over the centre O of the board.

Height-weighted balance is the row and column conditions. Now give each cell (x_{ij}, y_{ij}) of the grid the signed weight z_{ij}; the total weight is 0. Write $R_i = \sum_j z_{ij}$ and $C_j = \sum_i z_{ij}$ for the *row and column excesses*; by Lemma 2.2(3), S is semi-magic if and only if all R_i and C_j vanish. For every function $\varphi\colon \mathbb{R} \to \mathbb{R}$, grouping the cells by column and by row gives

$$\sum_{i,j} z_{ij}\,\varphi(x_{ij}) = \sum_j \varphi(x_j)\,C_j, \qquad \sum_{i,j} z_{ij}\,\varphi(y_{ij}) = \sum_i \varphi(y_i)\,R_i. \tag{3}$$

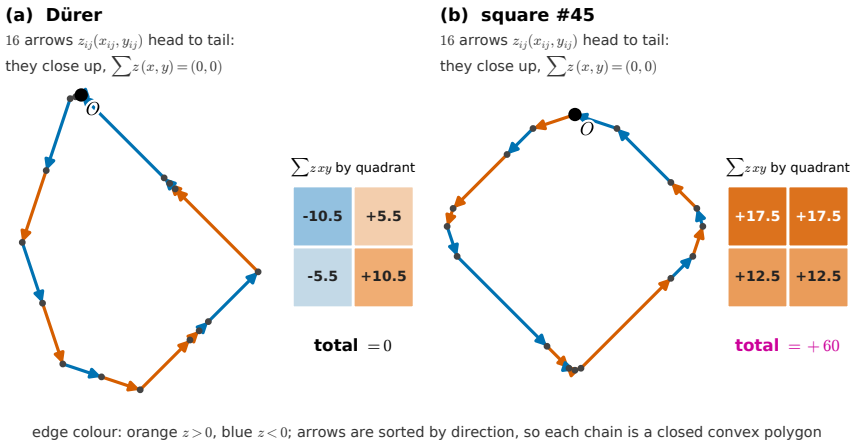


Figure 5: The dipole moment $\sum z_{ij}\,(x_{ij}, y_{ij})$ for Dürer's square and square #45: the 16 vectors $z_{ij}(x_{ij}, y_{ij})$ (orange for $z > 0$, blue for $z < 0$), sorted by direction and laid head to tail, close up at O. The tables give the mixed moment $\sum z\,xy$ over the four quadrants of the grid; the totals are 0 and 60.

Since the functions $\varphi(x)$ on the grid span the indicator functions of the columns, S is semi-magic if and only if both sums vanish for every φ. For $\varphi(t) = t$ this says that the first moments $\sum z_{ij}x_{ij}$ and $\sum z_{ij}y_{ij}$ of the heights vanish; Rogers and Loly call $\sum z\,(x, y)$ the *dipole moment* [31]. Since $\sum x_{ij} = \sum y_{ij} = 0$, the dipole moment is also the first moment $\sum s_{ij}(x_{ij}, y_{ij})$ of piles of s_{ij} checkers (Figure 4); Figure 5 shows the vectors $z_{ij}(x_{ij}, y_{ij})$ closing up. For $\varphi(t) = t^2$, (3) gives Loly's theorem [21]: $\sum_{i,j} s_{ij}(x_{ij}^2 + y_{ij}^2) = n^2(n^4 - 1)/12$ for every semi-magic square. The vanishing of the first moments is strictly weaker than the row and column conditions, and the diagonal conditions are not of the form (3): they do not determine the mixed moment $\sum z\,xy$, which is 0 for Dürer's square and 60 for square #45 (Figure 5; Appendix C).

Second moments. The second moments of the point set are as rigid as the first: when the first moments of the heights vanish they do not depend on the arrangement at all.

Lemma 3.1 (Covariance)**.** *The covariance matrix* $\Sigma = \frac{1}{N}\sum_{i,j} \mathbf{p}_{ij}\mathbf{p}_{ij}^{\top}$ *of the points of an arrangement* S *has* $\Sigma_{xx} = \Sigma_{yy} = a_n$, $\Sigma_{zz} = b_n$, $\Sigma_{xy} = 0$, $N\Sigma_{xz} = \sum_j x_j C_j$ *and* $N\Sigma_{yz} = \sum_i y_i R_i$. *If the first moments* $\sum z_{ij}x_{ij}$ *and* $\sum z_{ij}y_{ij}$ *vanish, for instance if* S *is semi-magic, then* $\Sigma = \operatorname{diag}(a_n, a_n, b_n)$, *and* $a_n < b_n$ *for* $n \geq 2$.

Proof. Lemma 2.2(1)–(2) and (3) with $\varphi(t) = t$. □

Since $b_n = (n^2 + 1)a_n$, placing equal point masses at the N points $\mathbf{p}_{ij}$ of an arrangement with vanishing first moments gives an axisymmetric inertia tensor $I = \operatorname{diag}(I_\perp, I_\perp, I_{zz})$ about $\mathbf{0}$ with $I_\perp/I_{zz} = (a_n + b_n)/(2a_n) = (n^2+2)/2$ (9 for order four), so in torque-free motion the point-weighted gem precesses like a symmetric rotor even when $\mathcal{G}(S)$ has no geometric symmetry.

4 Symmetry and congruence

4.1 Point sets of every order

Let $\tau(x, y, z) = (x, y, -z)$ be the reflection in the plane $z = 0$. The *complement* of an arrangement is $\overline{S} = N + 1 - S$, with height function $-z$; S is *self-complementary* if $\overline{S} = gS$ for some $g \in D_4$, and *associative* if $\overline{S} = r^2 S$, i.e. if entries in cells symmetric about the centre add up to $N + 1$. Complements of magic squares are magic. Since the points of $\overline{S}$ are $\tau\mathbf{p}_c$,

$$\mathcal{G}(\overline{S}) = \tau\,\mathcal{G}(S). \tag{4}$$

Lemma 4.1 (Elements of D_4 that can complement)**.** *Let* $\overline{S} = gS$ *for an arrangement* S *of order* $n \geq 2$ *and some* $g \in D_4$. *Then* g *is unique,* $g^2 = \mathrm{id}$, *and* g *fixes no cell except possibly the centre*

cell of an odd grid, whose entry is then $\bar{s}$. Consequently $g = r^2$ (so S is associative) if n is odd, and $g \in \{r^2, h, v\}$ if n is even. The map $\iota_g = \hat{g}\,\tau$ is an isometry of $\mathbb{R}^3$ that maps the point set of S onto itself, so $\iota_g \in \mathrm{Iso}(\mathcal{G}(S))$; explicitly $\iota_{r^2} = -\mathrm{id}$ (a point reflection), $\iota_h(x,y,z) = (-x,y,-z)$ and $\iota_v(x,y,z) = (x,-y,-z)$ (half-turns about the y- and the x-axis).

Proof. If also $\overline{S} = g'S$ then $g^{-1}g'$ fixes S; as the entries are distinct it fixes every cell, and D_4 acts faithfully on the cells, so $g = g'$. Since complementation commutes with g, $g^2S = g\overline{S} = \overline{gS} = \overline{\overline{S}} = S$, so $g^2 = \mathrm{id}$. If g fixes a cell c then $2s_c = N + 1 = 2\bar{s}$, so $s_c = \bar{s}$. The involutions of D_4 are r^2, h, v and the two diagonal reflections; the diagonal reflections fix $n \geq 2$ cells, h and v fix the n cells of the middle column or row when n is odd and none when n is even, and r^2 fixes only the centre cell when n is odd. Finally, the points of S are mapped by τ to those of $\overline{S} = gS$, and by $\hat{g} = \hat{g}^{-1}$ back to those of $g^2S = S$, by (2). □

Proposition 4.2 (Point sets with vanishing first moments)**.** *Let S and T be arrangements of order $n \geq 2$ with $\sum z_{ij}x_{ij} = \sum z_{ij}y_{ij} = 0$ (the first moments of the heights vanish), for instance semi-magic squares, and let P_S, P_T be their point sets.*

1. *Every isometry ψ of $\mathbb{R}^3$ with $\psi(P_S) = P_T$ is of the form $\hat{g}$ or $\hat{g}\,\tau$ with $g \in D_4$; $\hat{g}(P_S) = P_T$ if and only if $T = gS$, and $\hat{g}\,\tau(P_S) = P_T$ if and only if $T = g\overline{S}$.*
2. *P_S and P_T are congruent if and only if $T \in D_4S \cup D_4\overline{S}$.*
3. *$\mathrm{Iso}(P_S) = \{\mathrm{id}\}$ unless S is self-complementary, and $\mathrm{Iso}(P_S) = \{\mathrm{id}, \iota_g\}$ if $\overline{S} = gS$.*

Proof. An isometry of $\mathbb{R}^3$ mapping one finite set onto another maps centroid to centroid, and both centroids are $\mathbf{0}$ (Lemma 2.2(1)); so ψ is orthogonal, and $\psi\Sigma_S\psi^\top = \Sigma_T$ for the covariance matrices. By Lemma 3.1, $\Sigma_S = \Sigma_T = \mathrm{diag}(a_n, a_n, b_n)$ with $a_n \neq b_n$, so ψ commutes with Σ_S and preserves its eigenspaces, the plane $z = 0$ and the z-axis: $\psi = \mathrm{diag}(B, \varepsilon)$ with $B \in O(2)$ and $\varepsilon = \pm 1$. The projection of both point sets to the plane is the grid, so B is a symmetry of the grid, that is, an element $g \in D_4$; hence $\psi = \hat{g}$ or $\psi = \hat{g}\tau$. By (2), $\hat{g}(P_S) = P_{gS}$ and $\hat{g}\tau(P_S) = P_{g\overline{S}}$, and the point set determines the arrangement; this proves (1), and (2) follows. For (3) take $T = S$: $\hat{g}(P_S) = P_S$ forces $gS = S$, hence $g = \mathrm{id}$, and $\hat{g}\tau(P_S) = P_S$ forces $g\overline{S} = S$, i.e. $\overline{S} = g^{-1}S = gS$ by Lemma 4.1. □

The diagonal conditions play no role.

4.2 The isometry groups of the order-four gems

For the gem, $\mathrm{Iso}(\mathcal{G}(S)) = \mathrm{Iso}(\mathrm{vert}\,\mathcal{G}(S))$, since isometries preserve extreme points, and $\mathrm{Iso}(P_S) \subseteq \mathrm{Iso}(\mathcal{G}(S))$; the inclusion is an equality when every point is a vertex, but in general the gem might have isometries that do not preserve the non-vertex points. For order four this does not happen.

Proposition 4.3 (Isometry groups of the order-four gems; exact computation)**.** *For each of the 880 D_4-classes of magic squares S of order 4, $\mathrm{Iso}(\mathcal{G}(S)) = \mathrm{Iso}(P_S)$. Hence:*

1. *for the 48 associative squares, $\mathrm{Iso}(\mathcal{G}(S)) = \{\mathrm{id}, -\mathrm{id}\}$;*
2. *for the 304 self-complementary squares that are not associative, $\mathrm{Iso}(\mathcal{G}(S)) = \{\mathrm{id}, \iota_g\}$ with $g \in \{h, v\}$, a half-turn about a horizontal coordinate axis;*
3. *for the remaining 528 squares, $\mathrm{Iso}(\mathcal{G}(S))$ is trivial.*

In particular a gem has an improper isometry if and only if S is associative, and the 832 gems in cases (2) and (3) are chiral (chiral: having no orientation-reversing isometry).

Proof. The squares were sorted into the three cases by comparing $\overline{S}$ with the eight images gS; by Lemma 4.1 the complementing element is r^2, h or v. For each square we computed $\mathrm{Iso}(\mathrm{vert}\,\mathcal{G}(S))$ in exact integer arithmetic (Appendix A); its order is 2 in cases (1) and (2) and 1 in case (3), equal to the order of $\mathrm{Iso}(P_S)$ by Proposition 4.2(3), and $\mathrm{Iso}(P_S) \subseteq \mathrm{Iso}(\mathcal{G}(S))$. The maps ι_h, ι_v are proper. □

The partition (1)–(3) is classical: the associative squares form Dudeney's group III and the self-complementary non-associative ones his group VI [13]. What Proposition 4.3 adds is that the hulls have no further symmetry. Table 2 summarises the three cases and Figure 6 shows one gem of each.

Table 2: Symmetry types of the 880 order-four gems; "classes" counts congruence classes.

type of S	Dudeney	squares	$\mathrm{Iso}(\mathcal{G})$	classes	V
associative	III	48	$\{\pm\mathrm{id}\}$	48	$10, 12, 14, 16$
self-complementary, not associative	VI	304	$\{\mathrm{id}, \iota_g\}$, half-turn	304	$8, 10, 12, 14, 16$
not self-complementary	others	528	trivial	264	$8, \ldots, 16$
total		880		616	

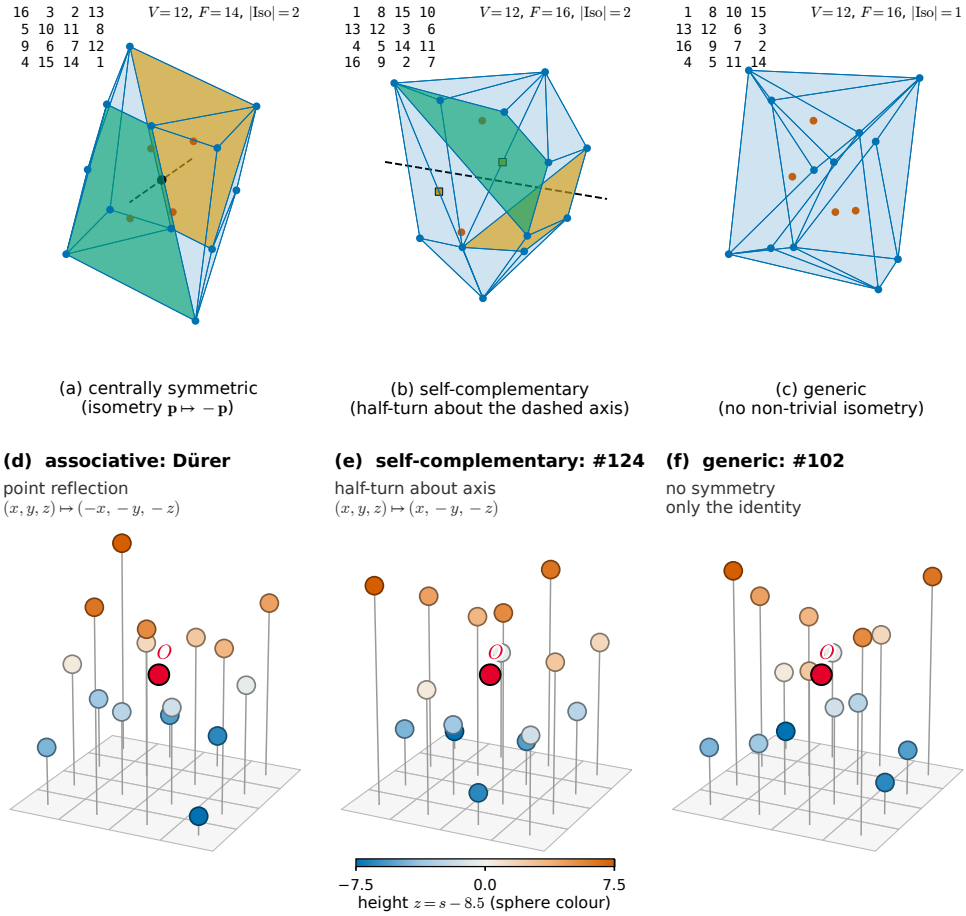


Figure 6: Gems (top) and point sets (bottom) of three order-four squares, all drawn from the same viewpoint, with the square printed above each gem. Interior points: red discs; non-vertex boundary points: orange squares. (a, d) Dürer's square, associative: a pair of opposite facets (orange, green), and the 8 antipodal pairs $\{\mathbf{p}, -\mathbf{p}\}$ joined through O. (b, e) Square #124, self-complementary and not associative: the half-turn about the dashed axis exchanges the highlighted facets and the 8 pairs of points. (c, f) Square #102: no non-trivial isometry; the gem is chiral and its mirror image in $z = 0$ is the gem of $17 - S$.

Lemma 4.4 (Parity)**.** *Let S be a self-complementary arrangement of order n that is not a D_4-image of the natural square (for instance, a self-complementary semi-magic square). Then V and*

V^∂ *are even and* $V^\circ \equiv n \pmod 2$. *If* S *is associative,* E *and* F *are even as well.*

Proof. Let $\overline{S} = gS$. Then $N + 1 - s_{gc} = s_c$, i.e. $z_{gc} = -z_c$, so $\iota_g \mathbf{p}_c = \mathbf{p}_{gc}$ for every cell c. The involution ι_g is an isometry of the 3-polytope $\mathcal{G}(S)$ (Proposition 2.3), so it permutes the vertices, the interior points and the non-vertex boundary points. By Lemma 4.1 it fixes $\mathbf{p}_c$ only if $gc = c$, i.e. only for the centre cell of an odd grid, whose point is the interior point $\mathbf{0}$. If S is associative, $\iota_g = -\mathrm{id}$, and a face fixed by $-\mathrm{id}$ would contain the centroid $\mathbf{0}$ of its vertices, which is interior; so $-\mathrm{id}$ pairs the edges and the facets. □

The hypothesis is needed: the natural square of order 3 is associative, but its gem is a parallelogram and has no interior points. For $n = 4$ the 48 associative gems have $V^\circ \in \{0, 2, 4\}$ (4, 23 and 21 gems), and among the 528 squares that are not self-complementary all four combinations of the parities of V and V° occur (`anc/verify_structure.py`).

Remark 4.5 (Facet orbits of the half-turns). The half-turn ι_h (the case ι_v is the same after the reflection d) maps the plane $\mathbf{a} \cdot \mathbf{p} = b$ to $\iota_h(\mathbf{a}) \cdot \mathbf{p} = b$. The two coincide if and only if $\iota_h(\mathbf{a}) = \mathbf{a}$, or $\iota_h(\mathbf{a}) = -\mathbf{a}$ and $b = 0$; a plane through the interior point $\mathbf{0}$ cannot support the gem. So an ι_h-invariant facet plane is one of the vertical planes $y = \pm\frac{n-1}{2}$ over the top and bottom rows, which by Proposition 5.2(2) below carry facets unless the row is an arithmetic progression, and F need not be even. Among the 304 order-four gems of case (2) of Proposition 4.3, 282 have two fixed facets, 18 one (so F is odd) and 4 none.

Remark 4.6 (Congruent facets). An isometry of a gem maps facets to congruent facets, so the facets of the 352 self-complementary gems of order four come in congruent pairs, apart from the facets fixed by a half-turn (Remark 4.5), none of which is congruent to another facet. These isometries are diagonal matrices, so the pairs stay congruent when the heights are rescaled, as in printed models and paper nets. Similarly, since $-\mathrm{id}\,\mathcal{G}(S) = \mathcal{G}(r^2\overline{S})$, every upper facet of $\mathcal{G}(S)$ is parallel and congruent to a lower facet of the half-turned complement $\mathcal{G}(r^2\overline{S})$, allowing copies of a gem and its complement twin to stack face-to-face into periodic columns; for instance, in square #762 the entries $\{13, 14, 15, 16\}$ lie on a centred knight's-move square of area 5 and are coplanar ($z = 6 - \frac{1}{5}(3x - y)$), forming a quadrilateral upper facet matched by $\{1, 2, 3, 4\}$ below. Congruence without symmetry is rare: of the 528 gems with trivial isometry group, 182 have congruent facets, 480 of their 8858 facets in 230 classes of triangles and 10 of quadrilaterals; with the heights scaled by 1/5 these numbers drop to 130 gems and 332 facets. The symmetric gems have congruences beyond their symmetry in 28 cases, and in 9 after scaling (`anc/congruent_facets.py`).

Remark 4.7 (Repeating chains of facets). The pairing of Remark 4.6 respects adjacency: an isometry of the gem maps facets that share an edge to facets that share an edge, so a chain of adjacent facets is matched by a chain of congruent facets in the same order. This is easiest to see on belts. If a plane through $\mathbf{0}$ contains no point $\mathbf{p}_c$, the facets that it crosses, taken in order along the boundary of the section of the gem by the plane, form a closed chain in which consecutive facets share an edge. If an isometry of the gem maps the plane to itself and acts on it as a half-turn about $\mathbf{0}$, it preserves the cyclic order of the edges of the section and fixes none of them, so it maps the chain to itself shifted half way round: the second half of the chain repeats the first. If it acts on the plane as a reflection, it reverses the chain. So for an associative square the chain along the plane $z = 0$ of the mean entry repeats after half a turn, and so does, for the other self-complementary squares, the chain along the vertical plane through $\mathbf{0}$ perpendicular to the axis of the half-turn; along $z = 0$ the half-turn reverses its chain, which then reads the same in both directions. For order four these chains have 8 or 10 facets for the associative gems, and 4 to 14 facets around the axis for the others (`anc/congruent_facets.py`).

5 Faces of the gem and a census of order four

After two facts about the faces of every gem, this section reports a computational census of the gems of the 880 magic squares of order four (vertices, faces, volumes, combinatorial types, congruence classes) and compares them with the gems of other arrangements.

5.1 Forced vertices and regular subdivisions

The *sides* of the grid are its first and last rows and columns. Every gem lies in the box $\mathcal{B}_n = [-\frac{n-1}{2}, \frac{n-1}{2}]^2 \times [-\frac{N-1}{2}, \frac{N-1}{2}]$ and meets each of its six faces: the top and the bottom face in the points of the entries N and 1, and the vertical face over a side e of the grid in the polygon $\operatorname{conv}\{\mathbf{p}_c : c \in e\}$. Each contact is a face of $\mathcal{G}(S)$, so its vertices are vertices of $\mathcal{G}(S)$; the lemma below names those that can be read off from the entries.

Lemma 5.1 (Forced vertices)**.** *Let S be an arrangement of order $n \geq 2$. The following points are vertices of $\mathcal{G}(S)$; we call them the* forced vertices *of S: the points over the four corner cells; the points carrying the entries 1 and N; and, for each side, the points over the cells of the smallest and of the largest entry on that side.*

Proof. A point that is the unique maximiser of a linear functional over the N points is a vertex. The entries are distinct, so $\mp z$ have unique maximisers where $s = 1$ and $s = N$. The functional $-x + y$ attains its maximum only at the top-left corner, and similarly for the other corners. On the left column $x = -\frac{n-1}{2}$, while $x \geq -\frac{n-1}{2} + 1$ elsewhere; for $\varepsilon = 1/N$ the functional $-x + \varepsilon z$ exceeds $\frac{n-1}{2} - \frac{1}{2}$ on the left column and is less than $\frac{n-1}{2} - \frac{1}{2}$ elsewhere, so its unique maximiser is the point of largest height on the left column; $-x - \varepsilon z$ gives the smallest, and the other sides follow by symmetry. □

For order four the forced vertices number between 4 and 13, and in general other points are vertices as well (Section 5.2).

Gems and regular subdivisions. The gem is two regular subdivisions of the grid, by z and by $-z$, glued along the boundary of the grid (with vertical facets over the sides where the entries are not in arithmetic progression); Figure 3e shows the two subdivisions for the Lo Shu. Let A be the $n \times n$ grid of cell positions. For a height function $\omega\colon A \to \mathbb{R}$, the *regular subdivision* $\mathcal{S}(\omega)$ of A is obtained by projecting to the plane the lower faces of $\operatorname{conv}\{(c, \omega(c))\}$, i.e. the faces whose outer normals have negative last coordinate; its cells are the sets of grid points whose lifts lie on a common lower face [11, Ch. 2], [17, Ch. 7]. Cells are marked: a cell is a set of grid points, and its convex hull may contain grid points that are not in it. $\mathcal{S}(\omega)$ does not change if ω is multiplied by a positive scalar or an affine function of (x, y) is added to it; the height functions inducing a given subdivision form a relatively open polyhedral cone, its *secondary cone*. By definition, $\mathcal{G}(S)$ is the lift of A by the height function z of S, and its upper faces are the reflections in $z = 0$ of the lower faces of the lift by $-z$.

Proposition 5.2 (Gems as pairs of regular subdivisions)**.** *Let S be an arrangement of order $n \geq 2$ that is not a D_4-image of the natural square.*

1. *Every facet of $\mathcal{G}(S)$ is a lower facet, an upper facet or a vertical facet. Projection to the plane maps the lower facets onto the maximal cells of $\mathcal{S}(z)$ and the upper facets onto the maximal cells of $\mathcal{S}(-z)$.*

2. *The vertical facets lie in the four planes $x = \pm\frac{n-1}{2}$, $y = \pm\frac{n-1}{2}$; the plane over a side of the grid contains a facet if and only if the entries along that side, in cell order, are not an arithmetic progression.*

3. *$\mathbf{p}_c$ is a vertex of $\mathcal{G}(S)$ if and only if c is a vertex of a cell of $\mathcal{S}(z)$ or of $\mathcal{S}(-z)$.*

4. *The face lattice of $\mathcal{G}(S)$, with vertices labelled by cells, is determined by the pair $(\mathcal{S}(z), \mathcal{S}(-z))$, hence by the secondary cones containing z and $-z$. The complement $N + 1 - S$ exchanges the two subdivisions.*

Proof. (1) is the definition of $\mathcal{S}(z)$ and, after reflecting in $z = 0$, of $\mathcal{S}(-z)$. (2) A vertical facet lies in a plane $\ell(x, y) = \max_A \ell$ for a linear form ℓ; the maximisers of ℓ on the grid are a corner

or the n cells of a side, and the face is 2-dimensional only in the second case and only if the lifts of the side are not collinear, i.e. if the heights along the side are not an affine function of the position. (3) The vertical line through a vertex $\mathbf{p}_c$ meets $\mathcal{G}(S)$ in a segment with endpoint $\mathbf{p}_c$, so $\mathbf{p}_c$ lies on a lower or an upper face, of which it is a vertex; conversely the vertices of a face are vertices of the polytope. (4) Let U and L be the upper and lower boundary functions of $\mathcal{G}(S)$ over $\operatorname{conv} A$. $U - L$ is concave, non-negative, and positive at the projection of an interior point, hence positive on $\operatorname{int}\operatorname{conv} A$. So a lower face and an upper face meet only over $\partial \operatorname{conv} A$. Over a side e of the grid, $\mathcal{G}(S)$ meets the vertical plane in $\operatorname{conv}\{\mathbf{p}_c : c \in e\}$. This polygon is bounded by the restrictions of the two boundary chains to e, and it is a vertical facet exactly when these chains differ, i.e. (by (2)) when the entries along e are not an arithmetic progression. A lower or upper facet projects bijectively onto its cell, so its vertices are the lifts of the vertices of the convex hull of the cell. Hence the marked cells of $\mathcal{S}(z)$ and $\mathcal{S}(-z)$, together with the vertical facets, determine every vertex–facet incidence. The complement has height function $-z$. □

Example 5.3 (Lo Shu, continuing Example 2.4)**.** Writing each grid point as its entry, $\mathcal{S}(z)$ has the maximal cells $\{1,2,6\}$, $\{1,3,8\}$, $\{1,2,3,4\}$, and $\mathcal{S}(-z)$ has $\{2,7,9\}$, $\{4,8,9\}$, $\{6,7,8,9\}$. The vertical facets are the triangles $\{2,4,9\}$, $\{2,6,7\}$, $\{1,6,8\}$, $\{3,4,8\}$. The centre point (height 0) lies in the convex hull of each quadrilateral cell but belongs to no cell: the plane of $\{1,2,3,4\}$ is $z = -x - y - 3$, which is $-3 < 0$ at the centre.

For the 880 order-four gems, 856 have four vertical facets, 20 have three and 4 have two; the height functions of the 7040 order-four magic squares induce 417 distinct regular subdivisions of the 4×4 grid up to D_4, and for 548 of the 880 classes $\mathcal{S}(z)$ is a triangulation (`anc/verify_structure.py`).

5.2 The census

There are 8 magic squares of order 3, all images of the Lo Shu under D_4, and 7,040 of order 4, forming 880 D_4-classes; the order-four squares were listed by Frénicle de Bessy [15] and have been recounted by computer [6]. Dudeney sorted them into twelve groups according to the pattern of complementary pairs $(a, 17-a)$ [13]; since the numbering of the groups differs between reproductions of his table, we list the labels used here, for all 7040 squares together with the number # of their class, in `anc/all7040_dudeney.tsv`. The number 275,305,224 of magic squares of order five up to D_4 is due to Schroeppel (1973) [16]; for order six see [29, 23, 37].

For each of the 880 squares we computed $\mathcal{G}(S)$ exactly (integer facet inequalities, rational volumes; Appendix A). Table 3 and Figure 7 summarise the results.

Table 3: The 880 order-four gems by number V of vertices: numbers of squares, of combinatorial types (face lattices up to isomorphism), of f-vectors and of distinct exact volumes, and the volume as a fraction of the box $\mathcal{B}_4$ ($\operatorname{vol}\mathcal{B}_4 = 135$). A volume can occur for several values of V, so the volume counts add up to more than 155. Bottom: numbers of gems with k interior points ($V^\circ = k$) and with k non-vertex boundary points ($V^\partial = k$).

V	8	9	10	11	12	13	14	15	16	all
squares	7	10	103	80	380	106	162	14	18	880
types	4	1	27	32	129	52	85	7	15	352
f-vectors	4	1	6	5	7	5	5	3	3	39
volumes	4	5	51	33	111	40	65	6	14	155
min of $\operatorname{vol}/\operatorname{vol}\mathcal{B}_4$	.622	.588	.481	.549	.430	.456	.452	.498	.449	.430
mean of $\operatorname{vol}/\operatorname{vol}\mathcal{B}_4$	.642	.623	.631	.630	.634	.614	.608	.560	.536	.623
max of $\operatorname{vol}/\operatorname{vol}\mathcal{B}_4$	.648	.667	.701	.699	.721	.688	.691	.615	.598	.721

k	0	1	2	3	4	5	6
gems with $V^\circ = k$	37	32	353	178	280	0	0
gems with $V^\partial = k$	357	204	257	40	21	0	1

Vertices, faces and volumes. Between 8 and all 16 points are vertices; $V = 12$ is the most frequent value (380 gems), and the mean is $269/22 \approx 12.23$. The 39 f-vectors range from $(8, 14, 8)$ (one gem) to $(16, 36, 22)$ (nine gems); the most frequent is $(12, 26, 16)$ (164 gems). The volume takes 155 distinct values, from 58 to 292/3, with mean 84.05; its denominators are $1, 2, 3, 6$, because for even n all coordinates are half-odd integers, so that $\mathcal{G}(S) + (\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ is a lattice polytope. The Ehrhart polynomials of these lattice polytopes take 554 distinct values (Appendix B).

Combinatorial types. Two gems have the same combinatorial type if their vertex–facet incidence graphs are isomorphic by an isomorphism that maps vertices to vertices and facets to facets. Testing pairs of incidence graphs for isomorphism gives 352 types (Appendix A). By (4) the gems of S and $\overline{S}$ are mirror images, so every type is a union of complement orbits; the 93 types with one member are self-complementary squares, and of the 201 types with two members, 177 are mirror pairs and 24 consist of two self-complementary squares. The numbers of types with $1, 2, 3, \ldots$ members are

squares per type	1	2	3	4	5	6	7	8	10	13	14	16	26	32
types	93	201	8	20	4	7	4	5	5	1	1	1	1	1

The largest type, with $f = (10, 22, 14)$ (twelve triangles and two quadrilaterals), has 32 members.

Proposition 5.4 (Congruence classes of the order-four gems; exact computation)**.** *Let S and T be magic squares of order four. Then $\mathcal{G}(S)$ and $\mathcal{G}(T)$ are congruent if and only if $T \in D_4 S \cup D_4 \overline{S}$. There are* 616 *congruence classes of gems:* 352 *self-complementary classes and* 264 *mirror pairs.*

Proof. "If" is (2) and (4). For "only if": the sorted multiset of pairwise squared distances of the vertex set is a congruence invariant, and it takes 616 distinct values over the 880 gems (exact computation). Gems in one complement orbit are congruent, and there are 616 orbits (352 self-complementary classes and 264 pairs). Hence the congruence classes are exactly the complement orbits. □

Remark 5.5 (The vertex set determines the square). If S, T are magic squares of order four and $\hat{g}\,\mathrm{vert}\,\mathcal{G}(S) = \mathrm{vert}\,\mathcal{G}(T)$ for some $g \in D_4$, then $T = gS$. This is linear algebra plus a finite check. By (2), gS and T have the same vertex set, so they have the same entries on the vertex cells and the same set U of other cells. The ten line sums are linear equations for the entries on U, with the 0/1 matrix A_U of incidences between lines and cells of U; if $\mathrm{rank}\, A_U = |U|$, these entries are determined, and $T = gS$. This holds for 871 of the 880 classes; the rank and everything below are invariant under D_4, which permutes the lines. For the other nine, #45, #46, #47, #49, #50, #54, #165, #166 and #672, we listed every magic square that has the entries of S on the vertex cells of S: for five of them S is the only one, and #45, #47, #50, #54 have 1, 1, 7 and 7 others (for #45 the square #109), but the gems of all these other squares have different vertex sets (`anc/vertex_determination.py`). Since $g^{-1}T$ has the vertex set of $\mathcal{G}(S)$, it is one of these squares, so $g^{-1}T = S$.

Forced vertices and named squares. For order four nothing beyond the forced vertices of Lemma 5.1 is a vertex of every gem: the only cells that carry a vertex in all 880 gems are the four corners, and the only entries that always label a vertex are 1 and 16. In eight gems the point of the entry 2 is a non-vertex boundary point, for instance for square #593 below. The gems of the Lo Shu, of Dürer's square and of the Siamese square of order 5 have 8 of 9, 12 of 16 and 12 of 25 points as vertices and volumes 18, 92 and 830/3. The gem of Dürer's square is centrally symmetric with twelve vertices; exchanging its first two rows and its last two rows gives square #802, whose gem has all sixteen points as vertices. The squares cited in Figures 5, 6 and 7 and in the text are

$$\#45 = \begin{pmatrix} 1 & 6 & 11 & 16 \\ 8 & 15 & 2 & 9 \\ 12 & 3 & 14 & 5 \\ 13 & 10 & 7 & 4 \end{pmatrix} \quad \#54 = \begin{pmatrix} 1 & 6 & 11 & 16 \\ 15 & 12 & 5 & 2 \\ 14 & 9 & 8 & 3 \\ 4 & 7 & 10 & 13 \end{pmatrix} \quad \#102 = \begin{pmatrix} 1 & 8 & 10 & 15 \\ 13 & 12 & 6 & 3 \\ 16 & 9 & 7 & 2 \\ 4 & 5 & 11 & 14 \end{pmatrix} \quad \#124 = \begin{pmatrix} 1 & 8 & 15 & 10 \\ 13 & 12 & 3 & 6 \\ 4 & 5 & 14 & 11 \\ 16 & 9 & 2 & 7 \end{pmatrix}$$

$$\#379 = \begin{pmatrix} 2 & 12 & 5 & 15 \\ 13 & 7 & 10 & 4 \\ 16 & 6 & 11 & 1 \\ 3 & 9 & 8 & 14 \end{pmatrix} \qquad \#593 = \begin{pmatrix} 4 & 1 & 16 & 13 \\ 5 & 15 & 2 & 12 \\ 14 & 8 & 9 & 3 \\ 11 & 10 & 7 & 6 \end{pmatrix} \qquad \#802 = \begin{pmatrix} 5 & 10 & 11 & 8 \\ 16 & 3 & 2 & 13 \\ 4 & 15 & 14 & 1 \\ 9 & 6 & 7 & 12 \end{pmatrix}$$

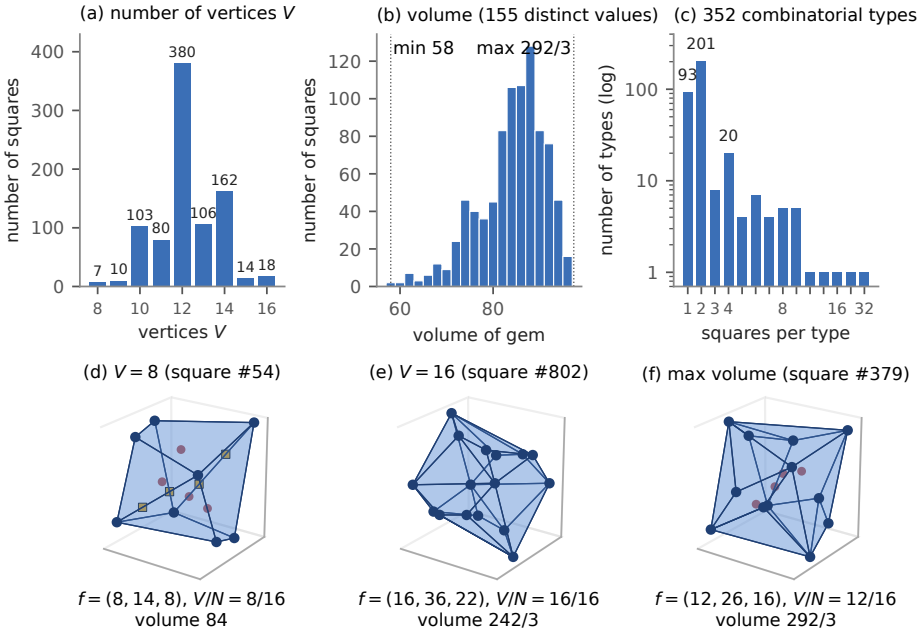


Figure 7: The 880 order-four gems. (a) Number of vertices. (b) Exact volume. (c) Sizes of the 352 combinatorial types (logarithmic scale). (d)–(f) Three extreme gems with translucent faces: the unique gem with $f = (8, 14, 8)$ (square #54; interior points as red discs, non-vertex boundary points as orange squares), a gem with all 16 points as vertices (square #802), and the gem of largest volume (square #379).

5.3 Which invariants see the line conditions

To see whether the gem distinguishes magic squares from other arrangements, we compared four nested families of order four: all $16! \approx 2.09 \cdot 10^{13}$ arrangements, the 6,697,177,424 arrangements with vanishing first moments (about one in 3124; `anc/energy_zeros_count.py`), the 549,504 semi-magic squares and the 7040 magic squares (Table 4; `anc/magic_vs_random.py`).

Table 4: Mean invariants of the gems of order four. The first three columns are estimates from 10^5 uniform samples each (standard errors at most 0.006 for the counts and the fractions, 0.024 for the volume and 0.001 for the centroid): arrangements, arrangements with $\sum z\,x = \sum z\,y = 0$ (rejection sampling), and semi-magic squares (uniform over all 549,504). The magic column is exact (all 7040 squares). d: difference of the means of magic and semi-magic squares divided by the pooled standard deviation, over all squares and over those that are not self-complementary (s.-c.); its standard error is at most 0.034 (0.044 in the last column). The solid centroid is the centre of mass of the solid $\mathcal{G}(S)$.

	random	first moments 0	semi-magic	magic	d, magic − semi-magic	
invariant	(estimates, 10^5 samples each)			(exact)	all	not s.-c.
vertices V	12.055	12.076	12.124	12.227	+0.08	+0.01
interior points V°	2.439	2.508	2.625	2.718	+0.10	+0.15
non-vertex boundary points V^∂	1.506	1.415	1.251	1.055	−0.20	−0.16
lower facets	6.503	6.503	6.455	6.439	−0.01	−0.03
volume	75.75	81.72	82.68	84.05	+0.21	+0.25
distance of the solid centroid from **0**	0.348	0.353	0.159	0.109	−0.62	−0.46
fraction with $\mathrm{Iso}(\mathcal{G}) \neq \{\mathrm{id}\}$	0.000	0.000	0.101	0.400	–	–

The counts barely move. The mean vertex count is 12.06 for random arrangements, 12.12 for semi-magic squares and 12.23 for magic squares ($d = 0.14$ against random arrangements, 0.08 against semi-magic squares). Self-complementary magic squares have more vertices on average (12.43, against 12.09 for the others; the 48 associative ones average 11.88), and they make up

40% of the magic squares but 10% of the semi-magic squares. This accounts for almost all of the excess over semi-magic squares: over the squares that are not self-complementary, magic and semi-magic squares differ by 0.017 ± 0.004 ($d = 0.01$). It does not account for the excess over random arrangements: the magic squares that are not self-complementary still exceed random arrangements by 0.040 ± 0.004, about a quarter of the excess. The mean numbers of lower and of upper facets differ by less than 0.08 between any two of the four families, and the mean number of vertical facets (3.97 to 4.00) by less than 0.03.

The continuous invariants respond to the hypotheses one at a time. Vanishing first moments turn the principal axes of the solid towards the coordinate axes (the tilt $(\Sigma_{xz}^2 + \Sigma_{yz}^2)^{1/2}$ of the covariance matrix Σ of the solid falls from 0.73 to 0.24, $d = -1.82$) and add 6.0 to the mean volume ($d = 0.90$), but do not move the solid centroid; the row and column conditions move it towards the origin (from 0.35 to 0.16, $d = -1.16$) and add 1.0 to the volume; and the diagonal conditions add another 1.4 to the volume ($d = 0.21$) and move the centroid further in ($d = -0.62$), also when the self-complementary squares are left out. Geometrically, $\operatorname{vol} \mathcal{G}(S) = \iint (U - L)$ for the upper concave and lower convex boundary functions U, L of Proposition 5.2(4): along every line L_0 with $\sum_{c \in L_0} z_c = 0$, convexity forces $\sum_{c \in L_0} L(c) \le 0 \le \sum_{c \in L_0} U(c)$, preventing z from being convex or concave along the line and pushing U and L apart. A non-trivial isometry occurs for 40% of the magic gems, 10% of the semi-magic gems and almost none of the others, in both families exactly for the self-complementary squares (Proposition 4.3; for the semi-magic sample by computation, consistent with Proposition 4.2).

At order five all four families are sampled, $2 \cdot 10^4$ each, and every hull is certified exactly; the semi-magic squares and the magic squares (distinct D_4-classes) come from the annealer of Appendix B, with other seeds than the larger magic sample of Table 7, whose hulls are computed in floating point. The distribution of the annealer at order five is not known; at order four it hits all 880 magic classes and reproduces the uniform distribution of the semi-magic squares (Appendix B). The mean vertex counts of random arrangements, arrangements with vanishing first moments, semi-magic and magic squares are 15.40, 15.42, 15.36 and 15.39 (standard errors 0.012; $|d| \le 0.04$; Table 7 gives 15.38 for its magic sample), the mean numbers of lower and of upper facets differ by at most 0.16, and all but 16 of the $8 \cdot 10^4$ gems have four vertical facets (the others have three). The volumes are 250.1, 260.9, 265.4 and 268.5 ($d = 0.22$ for magic against semi-magic squares), and the distance of the solid centroid from the origin is 0.47, 0.47, 0.24 and 0.22 ($d = -0.10$).

So the magic conditions hardly affect the vertex count; they show in the volume, the position of the solid and the frequency of symmetric squares, and beyond the row and column conditions only by a fraction of a standard deviation.

6 Larger orders: Siamese and linear squares

Beyond order four exhaustive computation is out of reach (275,305,224 magic squares of order five up to D_4), and samples suggest that the number of vertices grows with n (Table 7, estimates). The Siamese square behaves in the opposite way (Theorem 6.4, the main result); its proof uses a closed form, the invariance of $\mathcal{S}(\pm z)$ under adding affine functions (Section 5.1), the forced vertices and Proposition 4.2, and it also applies to the digit-exchanged square (Proposition 6.5); other linear squares have many more vertices (Remark 6.6).

6.1 Linear squares

The *Siamese* square of odd order $n = 2m + 1$ is de la Loubère's classical construction [10]: it has 1 in the middle cell of the top row; each next entry is placed one step up and one step to the right,

cyclically, or one step down if that cell is occupied. For $n = 5$ it is

$$\begin{pmatrix} 17 & 24 & 1 & 8 & 15 \\ 23 & 5 & 7 & 14 & 16 \\ 4 & 6 & 13 & 20 & 22 \\ 10 & 12 & 19 & 21 & 3 \\ 11 & 18 & 25 & 2 & 9 \end{pmatrix},$$

and for $n = 3$ it is a D_4-image of the Lo Shu. Write $\langle t \rangle \in \{0, \dots, n-1\}$ for the residue of t modulo n. The Siamese square is a *linear* (cyclic) square

$$s(c) = n\langle \mathbf{u} \cdot c + \alpha \rangle + \langle \mathbf{w} \cdot c + \beta \rangle + 1, \qquad c = (i, j), \quad \mathbf{u} = (u_1, u_2),\ \mathbf{w} = (w_1, w_2) \in \mathbb{Z}_n^2,\ \alpha, \beta \in \mathbb{Z}_n,$$

with $\mathbf{u} = (1, 1)$, $\mathbf{w} = (1, 2)$, $\alpha = -m$, $\beta = -2m$ (Step 1 of the proof of Theorem 6.4). If $c \mapsto (\langle \mathbf{u} \cdot c + \alpha \rangle, \langle \mathbf{w} \cdot c + \beta \rangle)$ is a bijection of $\mathbb{Z}_n^2$, the cell of the entry $nk_1 + k_0 + 1$ is an affine function of its digits (k_1, k_0), so the linear squares are the squares of the uniform step method: consecutive entries advance by a fixed step, and after every n entries by a fixed break-move [2, Ch. I]. When this method fills the square and when it gives equal row and column sums was analysed by Lehmer [20] (see [36]), and the method was studied further by Apostol and Zuckerman [3]; we give a self-contained proof of the following criterion, including the diagonals, and of the count.

Proposition 6.1 (Linear magic squares)**.** *Let n be odd. The linear square s is a magic square if and only if*

(a) u_1, u_2, w_1, w_2 and $u_1w_2 - u_2w_1$ are units of $\mathbb{Z}_n$, and

(b) for each digit, $(a_1, a_2, \gamma) = (u_1, u_2, \alpha)$ and (w_1, w_2, β), and each pair $(\sigma, \tau) = (a_1 + a_2, \gamma)$ (main diagonal) and $(a_1 - a_2, \gamma - a_2)$ (anti-diagonal): if $d = \gcd(\sigma, n) > 1$, then $\tau \equiv \frac{d-1}{2} \pmod d$.

Consequently there are $\varphi(n)^2 \sum f(r)f(r')$ linear magic squares of order n, the sum over the units r, r' of $\mathbb{Z}_n$ with $r - r'$ a unit, where $f(r) = n/(\gcd(r+1, n)\gcd(r-1, n))$, and one eighth as many D_4-classes. For a prime $n = p$ this is $(p-1)^3(p^3 - 6p^2 + 10p - 2)/8$ classes. If $3 \mid n$, then for every unit r one of $r \pm 1$ is divisible by 3, so that $f(r) \le n/3$.

Proof. Write $A(c) = \langle \mathbf{u} \cdot c + \alpha \rangle$ and $B(c) = \langle \mathbf{w} \cdot c + \beta \rangle$. The map $c \mapsto (A(c), B(c))$ is affine on $\mathbb{Z}_n^2$ with linear part of determinant $u_1w_2 - u_2w_1$, so s is a bijection onto $\{1, \dots, N\}$ if and only if the determinant is a unit. Along a line $c = c_0 + ke$, $k = 0, \dots, n-1$, the digit A takes the values $\langle \sigma k + \tau \rangle$ with $\sigma = \mathbf{u} \cdot e$ and $\tau = \mathbf{u} \cdot c_0 + \alpha$. If $d = \gcd(\sigma, n)$, these run d times through the n/d residues $t, t + d, \dots, t + n - d$, where $t \in \{0, \dots, d-1\}$ and $t \equiv \tau \pmod d$. So $\sum_k A = d\big(\frac{n}{d}t + d\binom{n/d}{2}\big) = \frac{n(n-1)}{2} + n\,e_A$ with $e_A = t - \frac{d-1}{2}$ and $|e_A| \le \frac{d-1}{2} < n$, and likewise $\sum_k B = \frac{n(n-1)}{2} + n\,e_B$. The line sum of s is $n\sum A + \sum B + n = \mu_n + n^2 e_A + n e_B$; it equals μ_n only if $n \mid e_B$, i.e. $e_B = 0$, and then $e_A = 0$. So a line has sum μ_n if and only if, for each digit, σ is a unit or $\tau \equiv \frac{d-1}{2} \pmod d$. For the row i, $e = (0, 1)$, $\sigma = u_2$ and $\tau = u_1 i + \alpha$; if $d = \gcd(u_2, n) > 1$, the condition for the rows $i = 0, 1$ forces $d \mid u_1$, and then d divides the determinant. So, given bijectivity, all rows have sum μ_n if and only if u_2 and w_2 are units, and all columns ($e = (1, 0)$, $\sigma = u_1$) if and only if u_1 and w_1 are units. The main diagonal has $c_0 = (0, 0)$, $e = (1, 1)$, and the anti-diagonal $c_0 = (0, n-1)$, $e = (1, -1)$, so $\tau \equiv \alpha - u_2$ there; this proves (a) and (b). For the count write $u_1 = ru_2$ and $w_1 = r'w_2$ with units r, r'; the determinant is $u_2w_2(r - r')$. With $d_1 = \gcd(r+1, n)$ and $d_2 = \gcd(r-1, n)$, (b) prescribes α modulo d_1 and modulo d_2; $\gcd(d_1, d_2)$ divides $\gcd(2, n) = 1$, so exactly $n/(d_1d_2) = f(r)$ values of α are admissible, and likewise $f(r')$ values of β. Summing over u_2, w_2, r, r' gives the count; D_4 acts freely on arrangements and maps linear magic squares to linear magic squares, hence the factor $1/8$. For $n = p$ prime, $f(r) = p$ unless $r = \pm 1$, and $f(\pm 1) = 1$; so $\sum_r f(r) = (p-1)(p-2)$, $\sum_r f(r)^2 = (p-3)p^2 + 2$, and the sum over $r \ne r'$ is $(p-1)^2(p-2)^2 - (p-3)p^2 - 2 = (p-1)(p^3 - 6p^2 + 10p - 2)$. If $3 \mid n$ and r is a unit, then $3 \nmid r$, so 3 divides $r - 1$ or $r + 1$. □

Corollary 6.2 (Two linear magic squares)**.** *For odd $n = 2m+1$, both*

$$s(i,j) = n\langle i+j-m\rangle + \langle i+2j-2m\rangle + 1 \quad \textit{and} \quad s'(i,j) = n\langle i+2j-2m\rangle + \langle i+j-m\rangle + 1$$

are magic squares. We call s' the digit-exchanged square*.*

Proof. The digits have $(a_1, a_2, \gamma) = (1, 1, -m)$ and $(1, 2, -2m)$; their coefficients are units, as n is odd, and the determinant is ± 1. For the first, the main diagonal has the unit step $\sigma = 2$, and the anti-diagonal $\sigma = 0$, $d = n$, $\tau = -m-1 \equiv m = \frac{n-1}{2}$. For the second, the anti-diagonal has $\sigma = -1$, and the main diagonal $\sigma = 3$, a unit unless $3 \mid n$; then $m \equiv 1 \pmod 3$ and $\tau = -2m \equiv 1 = \frac{3-1}{2} \pmod 3$. □

For $n = 5, 7, 11$ the formula gives 184, 3159 and 89,125 classes. For $n = 9$ the units $r \equiv 1 \pmod 3$ have $f(r) = 1, 3, 3$ ($r = 1, 4, 7$), those with $r \equiv 2$ have $f = 3, 3, 1$ ($r = 2, 5, 8$), and $r - r'$ is a unit if and only if $r \not\equiv r' \pmod 3$; so there are $6^2 \cdot 2 \cdot 7^2 = 3528$ linear magic squares and 441 classes. This explains the small number at $n = 9$: when $3 \mid n$ every linear magic square has, for each digit, a diagonal along which the digit is not a permutation of $\mathbb{Z}_n$, and the offsets are restricted accordingly. The script `anc/linear_magic_count.py` confirms the criterion and the count against the actual line sums of all n^6 parameter choices for $n = 3, 5, \ldots, 25$.

6.2 The Siamese theorem

Both proofs below end with the following certificate, which is also the one applied to every computed hull (Appendix A).

Lemma 6.3 (Closing a list of facets)**.** *Let $Q \subset \mathbb{R}^3$ be finite and $W \subseteq Q$, and let $\mathbf{a}_k \cdot \mathbf{p} \le b_k$ ($k = 1, \ldots, K$, $K \ge 2$, with pairwise distinct planes) hold at every point of Q. Suppose that each plane $\mathbf{a}_k \cdot \mathbf{p} = b_k$ contains three non-collinear points of W, and that every edge of each polygon $F_k = \operatorname{conv}\{\mathbf{w} \in W : \mathbf{a}_k \cdot \mathbf{w} = b_k\}$ is an edge of some other F_l. Then $\operatorname{conv} Q = \operatorname{conv} W = \{\mathbf{p} : \mathbf{a}_k \cdot \mathbf{p} \le b_k \text{ for all } k\}$, its facets are $F_1, \ldots, F_K$, and its vertices are the vertices of these polygons.*

Proof. Let $\Pi = \operatorname{conv} W$ and let H be the solution set of the inequalities, so $\Pi \subseteq \operatorname{conv} Q \subseteq H$. Each plane supports Π and meets it in the 2-dimensional face F_k; as there are $K \ge 2$ distinct such planes, Π is a 3-polytope and the F_k are distinct facets. An edge of F_k is an edge of Π and lies in exactly two facets of Π, F_k and, by hypothesis, some F_l; so the list is closed under adjacency across edges. The facet-adjacency graph of a 3-polytope is connected, so $F_1, \ldots, F_K$ are all the facets of Π, and $\Pi = H$. □

Theorem 6.4 (Siamese gems)**.** *Let S be the Siamese square of odd order $n = 2m+1 \ge 3$.*

(a) For $n = 3$ (the Lo Shu up to D_4) the vertices of $\mathcal{G}(S)$ are the eight boundary points, $f = (8, 16, 10)$ (Example 2.4), and $\operatorname{Iso}(\mathcal{G}(S)) = \{\pm\mathrm{id}\}$.

(b) For $n \ge 5$ the vertices of $\mathcal{G}(S)$ are exactly the twelve forced vertices of Lemma 5.1, the points over the cells

$$\begin{array}{lll} (0,0)\colon \bar{s}+m+2, & (0,m-1)\colon N-1, & (0,m)\colon 1, \\ (0,2m)\colon \bar{s}+m, & (m-1,0)\colon N-m, & (m,0)\colon m+2, \\ (m,2m)\colon N-m-1, & (m+1,2m)\colon m+1, & (2m,0)\colon \bar{s}-m, \\ (2m,m)\colon N, & (2m,m+1)\colon 2, & (2m,2m)\colon \bar{s}-m-2, \end{array}$$

listed with their entries; $f = (12, 26, 16)$, with 12 triangular and 4 quadrilateral facets; and $\operatorname{Iso}(\mathcal{G}(S)) = \operatorname{Iso}(P_S) = \{\pm\mathrm{id}\}$.

So only 12 of the n^2 points are vertices, and the combinatorial type of the gem does not depend on $n \ge 5$.

Proof. Step 1: closed form. We show that S is the square s of Corollary 6.2. The map $\kappa(i,j) = (\langle i+j-m\rangle, \langle i+2j-2m\rangle)$ is a bijection of $\mathbb{Z}_n^2$, and $\kappa(0,m) = (0,0)$. Write $k-1 = nk_1+k_0$ with $0 \le k_0 < n$; we prove by induction on k that the entry k is placed in the cell $\kappa^{-1}(k_1,k_0)$. This holds for $k=1$. Suppose it holds for $1,\dots,k$, with k in the cell $c = \kappa^{-1}(k_1,k_0)$; the occupied cells are $\kappa^{-1}(k_1',k_0')$ with $nk_1'+k_0' < k$. The up-right move $(i,j) \to (i-1,j+1)$ keeps $i+j$ and increases $i+2j$ by 1, so it leads to $\kappa^{-1}(k_1,\langle k_0+1\rangle)$. If $k_0 < n-1$, this cell is empty and is the cell prescribed for $k+1$. If $k_0 = n-1$, i.e. $n \mid k$, it is the cell $\kappa^{-1}(k_1,0)$ of the entry nk_1+1, so the move is blocked; the downward move $(i,j) \to (i+1,j)$ increases both $i+j$ and $i+2j$ by 1 and leads to the empty cell $\kappa^{-1}(k_1+1,0)$, the cell prescribed for $k+1 = n(k_1+1)+1$. By Corollary 6.2, S is magic. It is also associative: by the closed form, $s(i,j)+s(2m-i,2m-j) = N+1$.

Step 2: the forced vertices. In every row and every column the leading digit $\langle i+j-m\rangle$ takes each value in $\{0,\dots,n-1\}$ once, so the largest entry of a side is in the cell where this digit is $n-1$ and the smallest where it is 0. On the top row $i=0$ this gives $j = m-1$ with entry $n(n-1)+\langle -2\rangle+1 = N-1$, and $j=m$ with entry 1; on the bottom row $i=2m$, $j=m$ with entry $n(n-1)+n = N$, and $j=m+1$ with entry $\langle n+1\rangle+1 = 2$; on the left column $j=0$, $i=m-1$ with entry $n(n-1)+\langle -m-1\rangle+1 = N-m$, and $i=m$ with $\langle -m\rangle+1 = m+2$; on the right column $j=2m$, $i=m$ with entry $n(n-1)+\langle 3m\rangle+1 = N-m-1$, and $i=m+1$ with $\langle 3m+1\rangle+1 = m+1$. The corner entries follow in the same way. The entries 1 and N are among the side extremes, and for $m \ge 2$ the eight side cells are distinct and are not corners; so the forced vertices are the twelve listed points. For $m=1$ the side extremes are the corners and the midpoints of the sides.

Proof of (a). For $n=3$ the forced vertices are the eight boundary points, and the only other point is the interior point $\mathbf{0}$ (Example 2.4). An isometry of $\mathcal{G}(S)$ maps its vertex set onto itself and so fixes the centroid $\mathbf{0}$ of the vertex set; hence it preserves P_S, and $\mathrm{Iso}(P_S) = \{\pm\mathrm{id}\}$ by Proposition 4.2, since S is associative. From now on let $m \ge 2$. It remains to show that no point other than the twelve forced vertices is a vertex.

Step 3: shear to five levels. With $x = j-m$, $y = m-i$ and $\langle t\rangle = t - n\lfloor t/n\rfloor$ the closed form gives

$$z = \lambda(x,y) + n\,\zeta(x,y), \qquad \lambda(x,y) = n(x-y+m) + (2x-y+m) + 1 - \bar{s},$$
$$\zeta(x,y) = -n\left\lfloor \tfrac{\ell_1}{n}\right\rfloor - \left\lfloor \tfrac{\ell_2}{n}\right\rfloor, \qquad \ell_1 = x-y+m, \quad \ell_2 = 2x-y+m = \ell_1 + x.$$

The map $\Phi(x,y,z) = (x,y,(z-\lambda(x,y))/n)$ is a shear (an affine bijection that preserves vertical lines), so $\Phi(\mathcal{G}(S)) = \mathrm{conv}\{(x,y,\zeta(x,y))\}$ has the same face lattice and vertex cells as $\mathcal{G}(S)$; equivalently, $\mathcal{S}(\pm z) = \mathcal{S}(\pm\zeta)$. After the shear the height ζ takes five values, and its level sets are the lattice points of five polygons Λ_k (the *level polygons*). Indeed, for $|x|,|y| \le m$ we have $-m \le \ell_1 \le 3m$ and $-2m \le \ell_2 \le 4m$, so both floors lie in $\{-1,0,1\}$. If $\ell_1 < 0$, then $x \le y-m-1 \le -1$, so $\ell_2 < 0$ and $\zeta = n+1$. If $\ell_1 \ge n$, then $x \ge y+m+1 \ge 1$, so $\ell_2 \ge n$ and $\zeta = -(n+1)$. If $0 \le \ell_1 \le 2m$, then $\lfloor \ell_2/n\rfloor = -1,0,1$ according as $y \ge 2x+m+1$, $|y-2x| \le m$, or $y \le 2x-m-1$. Conversely, if $|y-2x| \le m$, then $\ell_1 < 0$ is impossible, since it would force $\ell_2 < 0$, i.e. $y \ge 2x+m+1$; likewise $\ell_1 \ge n$ would force $y \le 2x-m-1$. So, in the box $|x|,|y| \le m$ (Figure 8a): $\zeta = n+1$ on $\Lambda_{+2} = \{y \ge x+m+1\} = \mathrm{conv}\{(-m,m),(-1,m),(-m,1)\}$; $\zeta = 1$ on $\Lambda_{+1} = \{2x+m+1 \le y \le x+m\} = \mathrm{conv}\{(-m,0),(-m,1-m),(-1,m-1)\}$; $\zeta = 0$ on $\Lambda_0 = \{|y-2x| \le m\} = \mathrm{conv}\{(-m,-m),(0,m),(m,m),(0,-m)\}$; and $\zeta = -1$, $-(n+1)$ on $\Lambda_{-1} = -\Lambda_{+1}$, $\Lambda_{-2} = -\Lambda_{+2}$.

Step 4: the hull. Let $q_1,\dots,q_{12}$ be the images under Φ of the twelve forced vertices and $q_{13},\dots,q_{16}$ the remaining four vertices of the level polygons; their coordinates are in Table 5. Let H be the polyhedron defined by the eight inequalities below and their images $(1')$–$(8')$ under $-\mathrm{id}$. Inequality (k) is the plane through the first three points of its list below (its normal is the cross

product of two edge vectors):

$$(1)\ \zeta \le 2m+2, \qquad (2)\ y \le m, \qquad (3)\ -x \le m, \qquad (8)\ -2y+\zeta \le 2m,$$
$$(4)\ -2(m+1)x+(2m+1)y-m\zeta \le m(2m+1), \qquad (7)\ 2mx-y+m\zeta \le m(2m-1),$$
$$(5)\ 2(m+1)x-2(m+1)y+(2m+1)\zeta \le 2m(m+1),$$
$$(6)\ 2m(m+2)x-(2m^2+2m-1)y+m(2m+1)\zeta \le m(2m^2+2m-1).$$

Table 5 lists the slack (right-hand side minus left-hand side) of (1)–(8) at the sixteen points q_k. Every non-zero entry is a product of positive constants, of linear factors ($m-1$, m, $m+1$, $m+2$, $2m-1$, $2m+1$, $2m+3$, $4m-3$, $4m-1$), which are positive for $m \ge 2$, and of the quadratics $2m^2+2m+1$, $2m^2+2m-1$, $4m^2+2m-1$, $4m^2-2m+1$, $4m^2+m+1$, $4m^2-3m+1$, m^2+m+1. Each of these quadratics is positive at $m=1$ and has its minimum at some $m \le 3/8$, hence is positive for all $m \ge 1$. The set $\{q_1,\dots,q_{16}\}$ is centrally symmetric, so all sixteen inequalities hold at all sixteen points and hence, by convexity, at all points (x,y,ζ), each of which lies in a level polygon. The zeros of the table show that the planes (1)–(8) contain the following q_k ($k \le 12$), in cyclic order:

$$(1)\colon q_1q_2q_3, \quad (2)\colon q_1q_2q_8q_7, \quad (3)\colon q_1q_3q_5q_4, \quad (4)\colon q_1q_4q_7,$$
$$(5)\colon q_2q_3q_6, \quad (6)\colon q_2q_6q_9, \quad (7)\colon q_2q_8q_9, \quad (8)\colon q_3q_5q_6,$$

with the antipodal lists for (1′)–(8′) ($q_{10}=-q_1$, $q_{11}=-q_3$, $q_{12}=-q_2$, $q_9=-q_4$, $q_8=-q_5$, $q_7=-q_6$). All four points of each quadrilateral are forced vertices, hence in convex position, so only the cyclic order needs checking. In the plane $y=m$, written in coordinates (x,ζ), $q_1=(-m,2m+2)$, $q_2=(-1,2m+2)$, $q_8=(m,0)$ and $q_7=(0,0)$ form a trapezoid whose parallel sides q_1q_2 and q_8q_7 are traversed in opposite directions. In the plane $x=-m$, written in coordinates (y,ζ), the line through $q_5=(-m,0)$ and $q_1=(m,2m+2)$ has height $(m+1)(y+m)/m$. This is $m+1>1$ at $y=0$, so $q_4=(0,1)$ lies below the line, and $(m+1)^2/m<2m+2$ at $y=1$, so $q_3=(1,2m+2)$ lies above it. Hence q_1q_5 is a diagonal and $q_1q_3q_5q_4$ is the cyclic order. (The non-vertex point $q_{13}=(-m,1-m,1)$ lies on plane (3) inside this quadrilateral.) These sixteen polygons (12 triangles and the 4 quadrilaterals $(2),(3),(2'),(3')$) have 26 edges, each shared by exactly two of them. By Lemma 6.3, with Q the sheared points and $W=\{q_1,\dots,q_{12}\}$, $\Phi(\mathcal{G}(S))=H$ has exactly these sixteen facets and the vertices $q_1,\dots,q_{12}$; mapping back by Φ^{-1}, the gem has the twelve forced vertices and f-vector $(12,26,16)$.

Step 5: isometries. S is associative (Step 1), so $\mathrm{Iso}(P_S)=\{\pm\mathrm{id}\}$ by Proposition 4.2. An isometry of $\mathcal{G}(S)$ fixes the centroid $\mathbf{0}$ of its centrally symmetric vertex set and is orthogonal. Write $\tilde{q}_k=\Phi^{-1}(q_k)$ for the forced vertices in the original coordinates: $\tilde{q}_1=(-m,m,m+2)$, $\tilde{q}_2=(-1,m,2m^2+2m-1)$, $\tilde{q}_3=(-m,1,2m^2+m)$, $\tilde{q}_4=(-m,0,-2m^2-m+1)$, $\tilde{q}_5=(-m,-m,-m)$, $\tilde{q}_6=(0,-m,2m^2+2m)$, and the other six are their negatives. The squared norms $3m^2<3m^2+4m+4<4m^4+4m^3-2m^2-2m+1<4m^4+4m^3+2m^2+1<4m^4+8m^3+m^2-4m+2<4m^4+8m^3+5m^2$ of $\tilde{q}_5,\tilde{q}_1,\tilde{q}_4,\tilde{q}_3,\tilde{q}_2,\tilde{q}_6$ are distinct for $m \ge 2$, so an isometry maps each $\tilde{q}_k$ to $\pm\tilde{q}_k$. The inner products of $\tilde{q}_1$ with $\tilde{q}_2,\dots,\tilde{q}_6$ are $2m^3+7m^2+4m-2$, $m(2m^2+6m+3)$, $-(2m^3+4m^2+m-2)$, $-m(m+2)$, $m(2m^2+5m+4)$, all non-zero, so all signs agree with that of $\tilde{q}_1$, and the isometry is $\pm\mathrm{id}$. □

All entries of Table 5 and the closure of the facet list are verified by `anc/siamese_slacks.py` in exact polynomial arithmetic. As a cross-check, not part of the proof, the theorem agrees with certified exact hull computations for $n=3,5,\dots,41$ (`anc/verify_structure.py`). For $m=1$ several entries vanish and the four triangles degenerate.

6.3 The digit-exchanged square

The method of the proof applies to other linear squares; for the digit-exchanged square it gives the following.

Table 5: Proof of Theorem 6.4, Step 4: the sixteen vertices $q_1, \dots, q_{16}$ of the level polygons $\Lambda_{+2}, \Lambda_{+1}, \Lambda_0, \Lambda_{-1}, \Lambda_{-2}$ (groups of rows) and the slacks of the inequalities (1)–(8) at them, as polynomials in m; zeros in bold. All non-zero entries are positive for $m \geq 2$. The points $q_1, \dots, q_{12}$ are the images of the forced vertices; $q_{13}, \dots, q_{16}$ are not vertices of the gem.

point	(x, y, ζ)	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
q_1	$(-m, m, 2m+2)$	**0**	**0**	**0**	**0**	$2(m-1)(m+1)$	$2m(m-1)(m+2)$	$2m(m-1)$	$2(m-1)$
q_2	$(-1, m, 2m+2)$	**0**	**0**	$m-1$	$2(m-1)(m+1)$	**0**	**0**	**0**	$2(m-1)$
q_3	$(-m, 1, 2m+2)$	**0**	$m-1$	**0**	$(m-1)(2m+1)$	**0**	$(m-1)(2m+1)$	$(m-1)(2m-1)$	**0**
q_4	$(-m, 0, 1)$	$2m+1$	m	**0**	**0**	$4m^2+2m-1$	$2m(2m^2+2m-1)$	$2m(2m-1)$	$2m-1$
q_{13}	$(-m, 1-m, 1)$	$2m+1$	$2m-1$	**0**	$(m-1)(2m+1)$	$2m^2+2m+1$	$(m+1)(2m^2+2m-1)$	$4m^2-3m+1$	1
q_{14}	$(-1, m-1, 1)$	$2m+1$	1	$m-1$	$m-1$	$4m^2+2m-1$	$(m+1)(4m^2-2m+1)$	$(m+1)(2m-1)$	$4m-3$
q_5	$(-m, -m, 0)$	$2(m+1)$	$2m$	**0**	$2m^2$	$2m(m+1)$	$2m^2(m+2)$	$2m(2m-1)$	**0**
q_6	$(0, -m, 0)$	$2(m+1)$	$2m$	m	$2m(2m+1)$	**0**	**0**	$2m(m-1)$	**0**
q_7	$(0, m, 0)$	$2(m+1)$	**0**	m	**0**	$4m(m+1)$	$2m(2m^2+2m-1)$	$2m^2$	$4m$
q_8	$(m, m, 0)$	$2(m+1)$	**0**	$2m$	$2m(m+1)$	$2m(m+1)$	$2m(m-1)(m+1)$	**0**	$4m$
q_9	$(m, 0, -1)$	$2m+3$	m	$2m$	$2m(2m+1)$	$2m+1$	**0**	**0**	$2m+1$
q_{15}	$(m, m-1, -1)$	$2m+3$	1	$2m$	$(m+1)(2m+1)$	$2m^2+2m-1$	$(m-1)(2m^2+2m-1)$	$m-1$	$4m-1$
q_{16}	$(1, 1-m, -1)$	$2m+3$	$2m-1$	$m+1$	$4m^2+m+1$	$2m+1$	$(m-1)(2m+1)$	$(m-1)(2m-1)$	3
q_{10}	$(m, -m, -2m-2)$	$4(m+1)$	$2m$	$2m$	$2m(2m+1)$	$2(m+1)^2$	$2m(m^2+m+1)$	$2m^2$	$2(m+1)$
q_{11}	$(m, -1, -2m-2)$	$4(m+1)$	$m+1$	$2m$	$(m+1)(2m+1)$	$4m(m+1)$	$(m+1)(4m^2-2m+1)$	$(m+1)(2m-1)$	$4m$
q_{12}	$(1, -m, -2m-2)$	$4(m+1)$	$2m$	$m+1$	$2(m^2+m+1)$	$4m(m+1)$	$2m(2m^2+2m-1)$	$2m(2m-1)$	$2(m+1)$

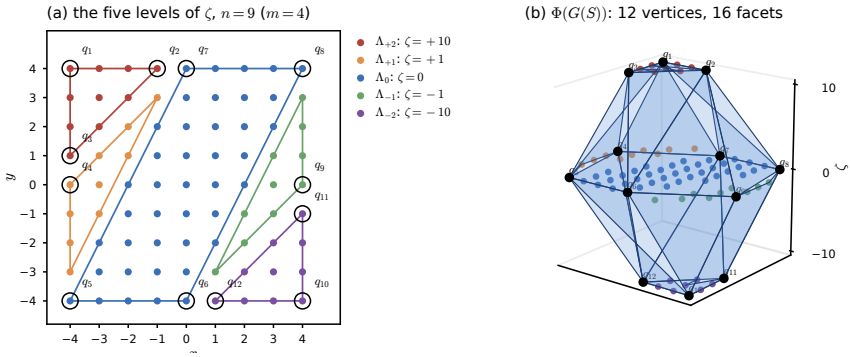


Figure 8: The Siamese gem of order $n = 9$ after the shear Φ. (a) The 81 cells coloured by the level value $\zeta \in \{\pm 10, \pm 1, 0\}$, with the five level polygons outlined and the twelve vertex cells $q_1, \dots, q_{12}$ circled. (b) $\Phi(\mathcal{G}(S))$ with its 12 vertices and 16 facets. The combinatorics of the level polygons does not depend on $m \geq 2$.

Proposition 6.5 (Digit-exchanged gems)**.** *Let S' be the digit-exchanged square s' of Corollary 6.2, of odd order $n = 2m+1 \geq 5$. The vertices of $\mathcal{G}(S')$ are the twelve points over the cells*

$$\begin{array}{llll} (0,0)\colon n+m+2, & (0,m-1)\colon N-n, & (0,m)\colon 1, & (0,2m)\colon N-n+m+1, \\ (1,2m)\colon m+2, & (m-1,0)\colon n(m+1), & (m+1,2m)\colon nm+1, & (2m-1,0)\colon N-n+m, \\ (2m,0)\colon m+1, & (2m,m)\colon N, & (2m,m+1)\colon n+1, & (2m,2m)\colon N-2n+m, \end{array}$$

listed with their entries; eight of them are forced vertices, and those over $(0, m-1)$, $(m-1, 0)$, $(m+1, 2m)$, $(2m, m+1)$ are not. The f-vector is $(12, 24, 14)$, with 8 triangular and 6 quadrilateral facets, and $\mathrm{Iso}(\mathcal{G}(S')) = \{\pm\mathrm{id}\}$.

Proof. Since $\langle 2m - t\rangle = n - 1 - \langle t\rangle$ and $(x, y) \mapsto (-x, -y)$ maps ℓ_1, ℓ_2 to $2m - \ell_1$, $2m - \ell_2$, the square S' is associative. In the coordinates of Step 3, $s' = n\langle \ell_2\rangle + \langle \ell_1\rangle + 1$, so $z = \lambda'(x, y) + n\zeta'$ with the affine function $\lambda' = n\ell_2 + \ell_1 + 1 - \bar{s}$ and $\zeta' = -n\lfloor \ell_2/n\rfloor - \lfloor \ell_1/n\rfloor$. The floors are those of Step 3, so ζ' is constant on the same level polygons, with the values $n+1$, n, 0, $-n$, $-(n+1)$ on $\Lambda_{+2}, \dots, \Lambda_{-2}$. Hence the shear $\Phi'(x, y, z) = (x, y, (z - \lambda')/n)$ maps $\mathcal{G}(S')$ onto the convex hull of the sixteen points q_k of Table 5 with the third coordinates ± 1 replaced by $\pm n$. Let H' be defined by the seven inequalities

$$\begin{array}{lll} (1)\ \zeta' \leq 2m+2, & (2)\ y \leq m, & (3)\ -x \leq m, \\ (4)\ -2(m+1)x + (m+1)y - m\zeta' \leq m(m+1), & & (5)\ x - y + m\zeta' \leq (m+1)(2m-1), \\ (7)\ -(2m+1)y + \zeta' \leq m(2m+1), & & \end{array} \qquad (6)\ 2x - y + \zeta' \leq m,$$

and their images (1′)–(7′) under −id. Table 6 (Appendix A) lists the slacks of (1)–(7) at the sixteen points. Every non-zero entry is a product of positive constants, of the linear factors $m-1$, m, $m+1$, $m+2$, $2m-1$, $2m+1$, $4m+3$ and of the quadratics $2m^2-1$, $2m^2+2m-1$, $2m^2+m+1$, $4m^2+m-1$, each of which is positive at $m=1$ and increasing for $m \geq 1$; so all are positive for $m \geq 2$, and the inequalities hold at all points as in Step 4. The zeros of the table give the facets

$$(1)\colon q_1q_2q_3, \quad (2)\colon q_1q_2q_8q_7, \quad (3)\colon q_1q_3q_{13}q_5, \quad (4)\colon q_1q_5q_7,$$
$$(5)\colon q_2q_3q_{13}, \quad (6)\colon q_2q_{13}q_6q_8, \quad (7)\colon q_{13}q_5q_6$$

and their antipodes ($q_{15} = -q_{13}$); the points q_4 and q_{14} lie on the planes (3) and (6) but are not vertices. The quadrilaterals are convex in the listed order: (2) is the trapezoid of Step 4; on the plane (3), in the coordinates (y, ζ'), the four turns $(b-a)\times(c-b)$ of consecutive vertices of $q_1q_3q_{13}q_5$ are $m-1$, $(m+1)(2m-1)$, $2(2m^2-1)$, $2(m^2-1)$; and (6) projects onto the quadrilateral $(-1,m)$, $(-m,1-m)$, $(0,-m)$, (m,m) of the grid, with turns $2m^2-1$, $m(2m+1)$, $2m(m+1)$, $(m+1)(2m-1)$. These fourteen polygons have 24 edges, each shared by exactly two of them, so by Lemma 6.3, with W the twelve points $\pm q_1, \pm q_2, \pm q_3, \pm q_5, \pm q_6, \pm q_{13}$ ($q_7 = -q_6$, $q_8 = -q_5$), they are the facets of $\Phi'(\mathcal{G}(S')) = H'$. Mapping back by $(i,j) = (m-y, x+m)$ gives the listed cells; the forced ones are the corners, $(0,m)$ and $(2m,m)$ (the entries 1 and N), $(1,2m)$ (the smallest entry of the right column) and $(2m-1,0)$ (the largest of the left column). For the isometries argue as in Step 5: the squared norms of the six antipodal pairs of vertices in the original coordinates are $2m^2+1$, $4m^4-4m^3-5m^2+4m+4$, $4m^4-3m^2+2$, $4m^4+4m^3-m^2-4m+2$, $m^2(4m^2+4m+3)$, $m^2(4m^2+8m+5)$ (over $(m-1,0)$, $(0,0)$, $(0,m-1)$, $(2m-1,0)$, $(2m,0)$, $(2m,m)$), pairwise distinct for $m \geq 2$, and the inner products of the vertex over $(0,0)$ with the other five are non-zero for $m \geq 2$. All of this is checked in exact polynomial arithmetic by `anc/digit_exchanged_slacks.py`, which applies the same certificate to the Siamese square as a cross-check. □

Remark 6.6 (Vertex counts of linear squares). The gems of other linear squares do not share the twelve-vertex count. Over all linear magic squares, counted up to D_4 (each family is closed under complementation; complements are not identified), the vertex counts range over $[11,18]$ for $n=5$ (184 classes), $[12,25]$ for $n=7$ (3159 classes), $[12,31]$ for $n=9$ (441 classes) and $[12,36]$ for $n=11$ (89,125 classes) (`anc/linear_squares.py`, exact hulls), and over $[12,41]$, $[12,38]$, $[12,48]$, $[12,51]$ and $[12,48]$ for $n=13,15,17,19,21$ (283,176, 1472, 1,713,664, 3,558,249 and 25,272 classes; `anc/linear_census.py`). For $n \geq 13$ the counts are floating-point (Quickhull) counts, and every class with at most 13 vertices or with the largest count is certified with the exact hull; for $n \leq 11$ the Quickhull vertex sets agree with the exact ones for every class. For $n=5$ the minimum 11 is attained by two classes, complements of each other, both pandiagonal and not associative, with f-vector $(11,23,14)$; one is $s = 5\langle 4i+3j\rangle + \langle 4i+2j\rangle + 1$ ($\mathbf{u}=(-1,-2)$, $\mathbf{w}=(-1,2)$, $\alpha=\beta=0$), the first square below.

$$\begin{pmatrix} 1 & 18 & 10 & 22 & 14 \\ 25 & 12 & 4 & 16 & 8 \\ 19 & 6 & 23 & 15 & 2 \\ 13 & 5 & 17 & 9 & 21 \\ 7 & 24 & 11 & 3 & 20 \end{pmatrix} \qquad \begin{pmatrix} 1 & 19 & 15 & 13 & 17 \\ 23 & 5 & 14 & 20 & 3 \\ 10 & 21 & 12 & 4 & 18 \\ 24 & 9 & 8 & 22 & 2 \\ 7 & 11 & 16 & 6 & 25 \end{pmatrix}$$

For $7 \leq n \leq 21$ the minimum 12 is attained by exactly two classes, those of the Siamese and the digit-exchanged square (Theorem 6.4, Proposition 6.5); for $n \geq 13$ this relies on the floating-point counts of the other classes. The Siamese gem is not vertex-minimal at order five: simulated annealing (Appendix B) found 15 D_4-classes of magic squares of order 5 whose gems have 10 vertices (`anc/sa_n5_V10.txt`). For example, the second square above has f-vector $(10,22,14)$, volume 845/3 and 8 interior points.

6.4 Fixed steps and the largest vertex count

The Siamese and the digit-exchanged gems have the same type for every odd $n \geq 5$. Every family of linear squares with fixed steps and offsets affine in n has an eventually periodic f-vector, while

the largest vertex count over all linear magic squares of order n is unbounded.

Fix $\mathbf{u}, \mathbf{w} \in \mathbb{Z}^2 \setminus \{0\}$ and rationals $\alpha_0, \alpha_1, \beta_0, \beta_1$, put $\alpha(n) = \alpha_0 + \alpha_1 n$, $\beta(n) = \beta_0 + \beta_1 n$ and $\mathcal{N} = \{n \geq 2 : \alpha(n), \beta(n) \in \mathbb{Z}\}$, and for $n \in \mathcal{N}$ let S_n be the array

$$s(c) = n\langle \mathbf{u} \cdot c + \alpha(n) \rangle + \langle \mathbf{w} \cdot c + \beta(n) \rangle + 1, \qquad c = (i, j) \in \{0, \ldots, n-1\}^2,$$

and $\mathcal{G}_n$ the convex hull of its lifted points $(x_j, y_i, s(c) - \frac{n^2+1}{2})$, the gem of S_n when S_n is an arrangement. Let $\delta \geq 1$ be the least integer with $\delta\alpha_1, \delta\beta_1 \in \mathbb{Z}$, and L the least common multiple of the non-zero numbers among $|u_1|$, $|u_2|$, $|w_1|$, $|w_2|$, $|u_1 w_2 - u_2 w_1|$. The Siamese squares form the family $\mathbf{u} = (1, 1)$, $\mathbf{w} = (1, 2)$, $\alpha(n) = -m = \frac{1-n}{2}$, $\beta(n) = -2m = 1 - n$, with $\mathcal{N}$ the odd numbers, $\delta = 2$ and $L = 2$.

Proposition 6.7 (Fixed steps). *(a) The number of vertices of $\mathcal{G}_n$ is bounded on $\mathcal{N}$.*

(b) There is an n_0 such that for every $n \in \mathcal{N}$ with $n \geq n_0$ and $\dim \mathcal{G}_n = 3$ also $n + \delta L \in \mathcal{N}$, $\dim \mathcal{G}_{n+\delta L} = 3$ and $f(\mathcal{G}_{n+\delta L}) = f(\mathcal{G}_n)$.

(c) If S_n is a semi-magic square, then $\dim \mathcal{G}_n = 3$.

Proof of (c), and outline of (a) and (b). If all the lifted points lay in a plane, it would not be vertical, as the grid points are not collinear; so $z = Ax + By + C$, equal row sums give $B = 0$, equal column sums $A = 0$, and then z would be constant. For (a) and (b), the proof of Theorem 6.4, Step 3, generalises: since $\langle t \rangle = t - n\lfloor t/n \rfloor$, one affine bijection of $\mathbb{R}^3$, the same for all cells, maps $\mathcal{G}_n$ onto the convex hull of the points $(c, \zeta(c))$ with $\zeta(c) = -n\lfloor t_A(c)/n \rfloor - \lfloor t_B(c)/n \rfloor$, where $t_A(c) = \mathbf{u} \cdot c + \alpha(n)$ and $t_B(c) = \mathbf{w} \cdot c + \beta(n)$. The pairs of floors range over a finite set that does not depend on n, and each level set is the set of lattice points of a polygon $\{x \in \mathbb{R}^2 : Mx \leq ng + h\}$ with M, g, h independent of n. For large n in a fixed residue class modulo δL, the vertices of the hull of the lattice points of such a polygon come from a fixed finite list of points $np + q$ (Lemma D.1), and the face lattice of the hull of finitely many points that move affinely with n is eventually constant (Lemma D.2). The details are in Appendix D. □

So the vertex count along such a family is bounded, as it is for the Siamese and the digit-exchanged squares, but the bound, n_0 and the eventual f-vectors are not explicit. The least period can be smaller than δL: for the Siamese family $\delta L = 4$ and the f-vector is constant for $n \geq 5$. In computations it can equal δL: for $\mathbf{u} = (1, 3)$, $\mathbf{w} = (3, 1)$, $\alpha = \beta = 0$ we have $\delta = 1$ and $L = 24$, and the exact hulls satisfy $f(\mathcal{G}_n) = f(\mathcal{G}_{n+24})$ for $150 \leq n \leq 197$, with no smaller shift doing so on that range (`anc/linear_fixed_steps.py`).

The bound depends on the family, and the largest vertex counts in Remark 6.6 grow with n. That they are unbounded follows from Rubin's example of triangles whose integer hulls have arbitrarily many vertices [33]. Zolotykh's survey [39, Sect. 3] describes Rubin's triangles by Fibonacci knapsack inequalities with $k + 3$ vertices; its printed inequality, with F_{2k-1} in place of F_{2k+1}, has fewer vertices, and we have not seen Rubin's paper, so we state and prove the version we use (Appendix D). Here F_t are the Fibonacci numbers, extended to all $t \in \mathbb{Z}$ by $F_0 = 0$, $F_1 = 1$, $F_{t+1} = F_t + F_{t-1}$.

Lemma 6.8 (Rubin's triangle). *Let $k \geq 1$, $a = F_{2k}$, $b = F_{2k+1}$, $R = b^2 - 1$ and $T_k = \{(x, y) \in \mathbb{R}^2 : x \geq 0,\ y \geq 0,\ ax + by \leq R\}$. The convex hull of $T_k \cap \mathbb{Z}^2$ has exactly $k + 3$ vertices, $(0, 0)$ and $(F_{2t},\ b - F_{2t-1})$ for $0 \leq t \leq k + 1$.*

Proposition 6.9 (Many vertices). *Let k, a, b, R be as in Lemma 6.8, and let n be a prime with $n > \max(b^2, 3b)$. Then $s(i, j) = n\langle ai + bj + n - 1 - R \rangle + \langle i + 3j \rangle + 1$ is a magic square of order n, and the points of its gem over the $k + 3$ cells (i, j) that are vertices of $\mathrm{conv}(T_k \cap \mathbb{Z}^2)$ are vertices of the gem.*

Proof. As $n > 3b \geq 6$ is prime, every integer σ with $0 < |\sigma| < n$ is a unit of $\mathbb{Z}_n$. The numbers a, b, 1, 3 and $3a - b = F_{2k} + F_{2k-2}$, and the diagonal steps $a + b = F_{2k+2}$, $a - b = -F_{2k-1}$, 4 and

-2, are of this kind, since $b^2 < n$ and $3b < n$; so (a) of Proposition 6.1 holds and (b) is vacuous, and the square is magic. Put $t_A = ai + bj + \alpha$ with $\alpha = n - 1 - R$, which lies in $\{1, \dots, n-1\}$ as $n \geq b^2 + 1$, and $t_B = i + 3j$. As in Step 3 of the proof of Theorem 6.4, $s - \frac{n^2+1}{2} = \lambda + n\zeta$ with $\lambda = nt_A + t_B + 1 - \frac{n^2+1}{2}$ affine and $\zeta = -n\lfloor t_A/n \rfloor - \lfloor t_B/n \rfloor$, so the shear $(x, y, z) \mapsto (x, y, (z - \lambda)/n)$ maps the gem onto the convex hull of the points $(x_j, y_i, \zeta(i, j))$. On the grid $t_A, t_B \geq 0$, so $\zeta \leq 0$, with equality if and only if $ai + bj \leq R$ and $i + 3j \leq n - 1$. Every lattice point (i, j) of T_k is a cell, as $i \leq R < n - 1$ and $j < b < n$, and satisfies $i + 3j \leq \max(R/a, 3R/b) < \max(b^2, 3b) < n$; so the cells with $\zeta = 0$ are exactly the lattice points of T_k. If $Q \subset \mathbb{R}^3$ is finite and Q_{top} is the set of its points of largest third coordinate, every vertex q of $\operatorname{conv} Q_{\text{top}}$ is a vertex of $\operatorname{conv} Q$: if a linear function ℓ is maximal on Q_{top} only at q, then $\zeta + \varepsilon\ell$ is maximal on Q only at q for small $\varepsilon > 0$. Applied to the sheared points, this proves the claim. □

Corollary 6.10. *For every prime $n \geq 7$ some linear magic square of order n has a gem with at least $\lceil \ln n/(4 \ln \varphi) \rceil + 2 \geq 0.519 \ln n + 2$ vertices, where $\varphi = (1 + \sqrt{5})/2$. In particular the largest vertex count of a linear magic square of prime order n is unbounded.*

Proof. Put $k = \lceil \ln n/(4 \ln \varphi) \rceil - 1$. Then $k \geq 1$, as $n \geq 7 > \varphi^4$, and $\varphi^{4k} < n$. Since $F_t \leq \varphi^{t-1}$ for $t \geq 1$, $b^2 = F_{2k+1}^2 \leq \varphi^{4k} < n$; and $3b < n$, as $3b = 6 < n$ for $k = 1$ and $3b < b^2$ for $k \geq 2$. Now apply Proposition 6.9. □

The bound is far below the computed maxima of Remark 6.6 (the square of the proof has 18 vertices at $n = 7$, where the bound is 4) and covers only prime orders, but its order of magnitude $\Theta(\ln n)$ is sharp: choosing balanced residues $u_1, u_2, w_1, w_2 \in (-n/2, n/2] \cap \mathbb{Z}$ and $\alpha, \beta \in \{0, \dots, n-1\}$ restricts the floor pair $(\lfloor t_A/n \rfloor, \lfloor t_B/n \rfloor)$ in Step 1 of Appendix D to at most 25 values on the grid, so the sheared points lie on at most 25 level polygons $P_{ab}(n) \subseteq [0, n-1]^2$ defined by at most 8 linear inequalities each, and the integer hull of any polygon with a bounded number of sides in $[0, n-1]^2$ has $O(\ln n)$ vertices [39, Sect. 2]. Lemma 6.8, Proposition 6.9 and Corollary 6.10 are checked by exact computation for $k \leq 15$, for $k \leq 3$ at $n = 7, 29, 173$, and at every prime $7 \leq n < 120$ (`anc/linear_fixed_steps.py`).

7 Open questions

1. *Hull versus point set.* Is $\operatorname{Iso}(\mathcal{G}(S)) = \operatorname{Iso}(P_S)$ for every magic, or semi-magic, square S, or more generally for every arrangement whose heights have vanishing first moments? By Proposition 4.2 this would determine the isometry groups of all these gems; it holds for order four (Proposition 4.3) and for the Siamese and digit-exchanged squares (Theorem 6.4, Proposition 6.5). For odd n the self-complementary squares are associative (Lemma 4.1), so the point set of an order-five square has isometry group $\{\pm\mathrm{id}\}$ or trivial; whether the same holds for the gems is open, and an exhaustive check over the 275,305,224 classes of order five would take about seven CPU-weeks at the rate of Section 5.2.

2. *Growth of the vertex count for arbitrary arrangements.* For uniformly random arrangements the mean vertex count grows from 8.22 at $n = 3$ to 27.91 at $n = 10$, and the ratio $\mathbb{E}[V]/\ln^2 N$ decreases slowly from 1.70 to 1.32 (Table 7; estimates). For N independent uniform points in a 3-polytope the expected vertex count is $\Theta(\ln^2 N)$ [4]; our points are not independent (the grid is fixed and only the heights are permuted), so this is only a heuristic analogy. Is $\mathbb{E}[V] = \Theta(\ln^2 N)$ for random arrangements? By Section 5.3 the line conditions hardly change the mean vertex count at orders four and five; do they change its growth rate?

3. *Secondary cones.* By Proposition 5.2 the face lattice of the gem of a height function z is fixed by the secondary cones containing z and $-z$, so the combinatorics of the gems of a family of arrangements is the set of pairs of chambers of the secondary fan of the grid that it meets. Which pairs are met by the permuted heights that satisfy the line conditions?

4. *Linear squares.* By Proposition 6.7 the vertex count is bounded along every family of linear squares with fixed steps and offsets affine in n, and the f-vector is eventually periodic, but neither the bound nor the onset of periodicity is explicit. Is the vertex count bounded uniformly in the offsets, for fixed steps? Over all linear magic squares of prime order n the largest vertex count is of order $\Theta(\ln n)$ (Corollary 6.10), with the lower bound $0.519 \ln n + 2$ far below the computed values (25, 36, 41, 48, 51 for $n = 7, 11, 13, 17, 19$; Remark 6.6). What is the asymptotic constant, and what happens for composite n?

5. *Inner core gems and solid mass balance.* While the N points always have centroid $\mathbf{0}$, the uniform solid $\mathcal{G}(S)$ does so for only 51 of the 880 order-four gems (Appendix C), because the outer hull ignores the $N - V$ non-vertex points. Peeling $\operatorname{vert} \mathcal{G}(S)$ leaves the inner core gem $\mathcal{G}_1(S) = \operatorname{conv}(P_S \setminus \operatorname{vert} \mathcal{G}(S))$, whose point set has centroid $-V/(N - V)$ times the vertex centroid of $\mathcal{G}(S)$. For order four $\mathcal{G}_1(S)$ is 3-dimensional for 511 of the 880 gems (207 of which already have solid centroid $\mathbf{0}$, and 33 of the half-turn gems have outer and inner solid centroids on opposite rays from $\mathbf{0}$, allowing a two-density composite solid to balance at $\mathbf{0}$). How many convex layers arise for larger orders?

Use of AI systems

The author conceived the Magic Gem construction in a spoken conversation with a chatbot (OpenAI's ChatGPT), used as a sounding board; the transcript is available from the author. All versions of this paper were prepared with substantial help from generative AI, which earlier versions did not state. Versions 1–4 were prepared with a mix of OpenAI's ChatGPT, Anthropic's Claude (Sonnet and Opus, used in the Cursor editor) and Google's Gemini, used for discussion, code and tests over the three years of the project. This version was restructured and largely rewritten with Anthropic's Claude, used through the agentic tool Claude Code in October 2026, which wrote and ran the code in `anc/`, proposed and wrote out proofs, and drafted the text; separate AI sessions acted as adversarial referees. Several errors in earlier versions were found in that process and are corrected here. The author directed the work and is responsible for the content; AI systems are not authors. Every number in the paper is recomputed by the scripts in `anc/`, the order-4 results by exact enumeration of all 880 squares. The author has checked every proof in Sections 2–6 and the appendices by hand.

Changes from earlier versions. Versions 1–4 contained the following errors and overstatements, which are corrected or withdrawn here. (i) The "Complete Energy Characterization Theorem" presented in version 4 as the main theoretical contribution is immediate from the definitions: the energy is a sum of squared line-sum deviations, so it vanishes exactly on magic squares. It is not presented as a result here. (ii) Table 2 of version 4 gave about 3.4×10^7 essentially different magic squares of order five; the correct number is 275,305,224 [16], and 3.4×10^7 is that number divided by 8. (iii) The same table gave 8, 12 and 16 as the "typical" numbers of hull vertices for orders 3, 4 and 5. For order four the number ranges from 8 to 16 (Table 3 of this version), and the Siamese gem of order five has 12 vertices (Theorem 6.4). (iv) The energy-landscape statistics of version 4 (its Section 4.5, including the "landscape entropy" and the claim that it is scale-invariant) are not carried over. (v) The statistics in Table 3 of version 4 weighted the diagonal terms of the energy differently from its stated definition. That table also reported that 6×10^7 uniformly random arrangements of order four contained no zero of the four-term energy other than magic squares; the zeros are the arrangements whose first moments vanish and whose two diagonals sum to 34. An exact count gives 14,639,440 such zeros among the 16! arrangements, 7,040 of them magic (`anc/energy_zeros_count.py`), so about 42 non-magic zeros were to be expected, and a rerun with the same number of samples finds 45 (`anc/v4_sampling_recheck.py`). That statement of version 4 is withdrawn. (vi) The hardware stated in version 4 was incorrect; the computations were run on an Apple M4 Max.

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A Computational methods

All computations use Python 3.9 with NumPy 2.0, SciPy 1.13 (Quickhull), SymPy 1.13 and NetworkX 3.2; the scripts and data are the ancillary files in `anc/`, whose `README` lists the commands.

Exact hulls. After doubling, all coordinates are integers. A floating-point Quickhull triangulation only proposes candidate facet planes, recomputed as primitive integer equations through three points each; the hull is then certified in integer arithmetic: (i) the planes are pairwise distinct; (ii) every point satisfies every facet inequality; (iii) the points on each plane have an exact 2-dimensional convex hull (monotone chain with integer cross products after an injective projection), the facet polygon; (iv) every edge of every facet polygon lies in exactly two facets. By Lemma 6.3, (i)–(iv) certify the hull; strict convexity across every edge and $V - E + F = 2$ are checked as redundant tests. An independent implementation that enumerates the supporting planes through all triples of points, without Quickhull, gives the same vertex sets, f-vectors and volumes for all 880 order-four gems, the Siamese squares of orders 3 to 21 and the order-five squares of Remark 6.6. For the Siamese family the script that generates Table 5 checks these conditions symbolically for every

$m \geq 2$. This certificate is applied to every exact hull in the paper (order four, Siamese, digit-exchanged and linear squares, the 15 ten-vertex order-five squares, and the samples of Section 5.3). Vertices are the points on at least three facet planes, interior points are those satisfying every facet inequality strictly, and volumes are exact rationals. For the 880 order-four gems this takes about 15 s in total.

Combinatorial types, isometries, congruence. Combinatorial types are the isomorphism classes of the bipartite vertex–facet incidence graphs, under isomorphisms that map vertices to vertices and facets to facets (so a polytope and its dual are not identified); candidates are grouped by a Weisfeiler–Lehman hash and confirmed by explicit isomorphism tests. Isometry groups of a finite set X (a vertex set or a full point set) are computed exactly: X is centred at its centroid, every isometry is then orthogonal and determined by the images of three linearly independent points, which are searched among the points with the same squared norms and mutual inner products, and a candidate is accepted only if the induced permutation of X preserves all pairwise integer squared distances. Congruence of two finite sets is tested by comparing the sorted multisets of squared distances and then by an isomorphism test of the complete graphs with squared distances as edge labels; for the 880 gems the multisets alone certify the classes (Proposition 5.4). Proposition 4.2 is checked in the same way for all 880 squares and for the Siamese squares of orders 3 to 15.

Polynomial identities. The slack polynomials of Table 5 are computed with SymPy; a polynomial p is certified positive for all integers $m \geq 2$ by non-negativity of all coefficients of $p(2+t)$ and $p(2) > 0$. The same script checks the closure of the facet list and the norms and inner products of Step 5. For the digit-exchanged square, `anc/digit_exchanged_slacks.py` applies the same certificate, written once for both squares (it reproduces Theorem 6.4 as a cross-check), to the slacks of Table 6; it also checks, for $n = 5, \ldots, 61$, that the sheared points of s' have the five level sets used in the proof of Proposition 6.5 and satisfy all fourteen inequalities.

Table 6: Proof of Proposition 6.5: the sixteen vertices $q_1, \ldots, q_{16}$ of the level polygons of the digit-exchanged square after the shear Φ' (heights $n+1$, n, 0, $-n$, $-(n+1)$) and the slacks of the inequalities (1)–(7) at them, as polynomials in m; zeros in bold. All non-zero entries are positive for $m \geq 2$. The points q_4, q_{14}, q_9, q_{16} are not vertices of the gem.

point	(x, y, ζ')	(1)	(2)	(3)	(4)	(5)	(6)	(7)
q_1	$(-m, m, 2m+2)$	**0**	**0**	**0**	**0**	$m-1$	$2(m-1)$	$2(2m^2-1)$
q_2	$(-1, m, 2m+2)$	**0**	**0**	$m-1$	$2(m-1)(m+1)$	**0**	**0**	$2(2m^2-1)$
q_3	$(-m, 1, 2m+2)$	**0**	$m-1$	**0**	$(m-1)(m+1)$	**0**	$m-1$	$(m+1)(2m-1)$
q_4	$(-m, 0, 2m+1)$	1	m	**0**	m^2	$m-1$	$m-1$	$(m-1)(2m+1)$
q_{13}	$(-m, 1-m, 2m+1)$	1	$2m-1$	**0**	$2m^2-1$	**0**	**0**	**0**
q_{14}	$(-1, m-1, 2m+1)$	1	1	$m-1$	$2m^2-1$	$m-1$	**0**	$2(m-1)(2m+1)$
q_5	$(-m, -m, 0)$	$2(m+1)$	$2m$	**0**	**0**	$(m+1)(2m-1)$	$2m$	**0**
q_6	$(0, -m, 0)$	$2(m+1)$	$2m$	m	$2m(m+1)$	$2m^2-1$	**0**	**0**
q_7	$(0, m, 0)$	$2(m+1)$	**0**	m	**0**	$2m^2+2m-1$	$2m$	$2m(2m+1)$
q_8	$(m, m, 0)$	$2(m+1)$	**0**	$2m$	$2m(m+1)$	$(m+1)(2m-1)$	**0**	$2m(2m+1)$
q_9	$(m, 0, -2m-1)$	$4m+3$	m	$2m$	$m(m+2)$	$4m^2+m-1$	$m+1$	$(m+1)(2m+1)$
q_{15}	$(m, m-1, -2m-1)$	$4m+3$	1	$2m$	$2m+1$	$2(m+1)(2m-1)$	$2m$	$2m(2m+1)$
q_{16}	$(1, 1-m, -2m-1)$	$4m+3$	$2m-1$	$m+1$	$2m+1$	$4m^2+m-1$	$2m$	$2(2m+1)$
q_{10}	$(m, -m, -2m-2)$	$4(m+1)$	$2m$	$2m$	$2m(m+1)$	$4m^2+m-1$	2	$2(m+1)$
q_{11}	$(m, -1, -2m-2)$	$4(m+1)$	$m+1$	$2m$	$(m+1)^2$	$2(m+1)(2m-1)$	$m+1$	$2m^2+m+1$
q_{12}	$(1, -m, -2m-2)$	$4(m+1)$	$2m$	$m+1$	$2(m+1)$	$2(m+1)(2m-1)$	$2m$	$2(m+1)$

B Lattice-point counts and vertex statistics

Ehrhart polynomials of the order-four gems. For even n all coordinates of every $\mathbf{p}_{ij}$ are half-odd integers, so $Q(S) = \mathcal{G}(S) + (\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ is a lattice polytope, with Ehrhart polynomial $L_S(t) = \#(tQ \cap \mathbb{Z}^3)$; it is the same for every lattice translate of Q and for all images gS, $g \in D_4$. For a lattice 3-polytope, $L_S(t) = c_3t^3 + c_2t^2 + c_1t + 1$, where c_3 is the volume and c_2 is half the sum of the areas of the facets, each measured relative to the lattice in its plane [5, Ch. 3 and 5]; and $L_S(1)$ is the number of lattice points of Q. So L_S is determined by the volume, this surface term

Table 7: Mean number of vertices of the gem. Top: exact values (all 8 and 7040 magic squares of orders 3 and 4; the Siamese squares). Bottom: estimates from samples. Random: uniformly random arrangements, 10^5 samples per order (standard errors at most 0.010). Magic, annealed: 59,993, 24,000, 8,000 and 2,400 distinct D_4-classes for $n = 5, \dots, 8$ (standard errors 0.007, 0.013, 0.025, 0.054), from a sampler whose distribution is not known; hulls in floating point. The certified order-five samples of Section 5.3 are separate.

n	3	4	5	6	7	8	9	10
exact								
magic, all squares	8	12.23	–	–	–	–	–	–
Siamese	8	–	12	–	12	–	12	–
estimates								
random arrangements	8.22	12.06	15.39	18.39	21.08	23.53	25.79	27.91
magic, annealed	–	–	15.38	18.41	21.08	23.56	–	–

and the number of lattice points. Over the 880 gems (`anc/ehrhart_880.py`) there are 554 distinct polynomials, against 616 congruence classes and 155 volumes; so this count only measures how often non-congruent gems share these three numbers. For the Lo Shu square (odd order: the gem itself is a lattice polytope) $L(t) = 1+t+13t^2+18t^3$, and for Dürer's square $L(t) = 1-3t+44t^2+92t^3$.

Vertex counts for larger orders (numerical). In Table 7 the magic squares of orders 5 to 8 are found by simulated annealing on the sum of squared line excesses (`anc/sa_magic.c`). For random arrangements the fraction $\mathbb{E}[V]/N$ falls from 0.91 ($n = 3$) to 0.28 ($n = 10$). For $n = 5, \dots, 8$ the annealed means are within 1.2 standard errors of the random ones; this is a statement about the samples only, as the sampling distribution of the annealer for $n \geq 5$ is not known (compare [29]). At $n = 4$, where all magic squares are known, $2 \cdot 10^4$ annealed squares hit all 880 classes, each between 10 and 41 times (χ^2 test against the uniform distribution: $p = 0.078$; `anc/vertex_stats_sa4.json`), and the annealer for semi-magic squares reproduces the uniform distribution of every invariant of Table 4 within two standard errors (`anc/magic_vs_random.log`). The smallest annealed vertex counts are 10, 12, 14 and 16 for $n = 5, \dots, 8$ (Remark 6.6). The hulls of these samples are computed by Qhull in floating point, with the vertex sets of the first 100 hulls of each batch rechecked in exact arithmetic; the seeds are $7000n + c$ for random arrangements (`anc/vertex_statistics.py`) and $1000s + n$, $s = 1, \dots, 16$, for the annealer (`anc/run_sa.sh`). The order-five samples of Section 5.3 were drawn separately ($2 \cdot 10^4$ per family, annealer seeds $9000s + 5$, $s = 1, \dots, 8$, every hull certified exactly); their magic sample has mean vertex count 15.39, against 15.38 here.

C Further order-four data

Solid centroid, inner core gem and mixed moment. The centre of mass of the solid $\mathcal{G}(S)$ is not the origin for 829 of the 880 gems; the other 51 include the 48 associative gems (`anc/solid_centroid.py`). Removing $\operatorname{vert} \mathcal{G}(S)$ leaves $N - V$ non-vertex points, which are non-empty for 862 of the 880 gems and span a 3-dimensional inner core gem $\mathcal{G}_1(S) = \operatorname{conv}(P_S \setminus \operatorname{vert} \mathcal{G}(S))$ for 511 of them (2-dimensional for 175, 1-dimensional for 162 and a single point for 14); for 207 of the 511 the solid centroid of $\mathcal{G}_1(S)$ is the origin. The mixed moment $|\sum z\,xy|$ takes 48 distinct values between 0 and 60 and vanishes for 56 classes (tables in Figure 5), among them the 48 associative ones, for which z is odd under $(x, y) \mapsto (-x, -y)$ and xy is even.

Difference-body ratio. For a convex body $K \subset \mathbb{R}^3$ with non-empty interior let $\sigma(K) = \operatorname{vol}(K - K)/(8 \operatorname{vol} K)$. By the Brunn–Minkowski inequality [34], $\sigma(K) \geq 1$, with equality if and only if K is centrally symmetric; by the Rogers–Shephard inequality [32], $\sigma(K) \leq \binom{6}{3}/8 = \frac{5}{2}$, with equality if and only if K is a simplex; see [18, 34] for other measures of symmetry. By Proposition 4.3, $\sigma = 1$ exactly for the 48 associative gems; for the other 832, $\sigma \in [1057/1018, 145/88] \approx [1.038, 1.648]$ (exact, `anc/difference_body_ratio.py`).

Dudeney's groups. Each of Dudeney's groups with more than eight members has gems with at least three vertex counts and 20 combinatorial types; the 8-vertex gems lie in groups VI (three) and VIII (four) (`anc/verify_numbers.py`). The 48 pandiagonal squares (group I [27]) have 11 to 15 vertices and one to four interior points, and none is associative.

D Proofs for Section 6.4

We use the notation of Section 6.4; $c = (i, j)$ runs over the cells, $t_A(c) = \mathbf{u} \cdot c + \alpha(n)$ and $t_B(c) = \mathbf{w} \cdot c + \beta(n)$ are integers for $n \in \mathcal{N}$, and f denotes the f-vector.

Step 1: levels. Since $\langle t \rangle = t - n\lfloor t/n \rfloor$, we have $s(c) - \frac{n^2+1}{2} = \lambda(c) + n\zeta(c)$ with $\lambda(c) = nt_A(c) + t_B(c) + 1 - \frac{n^2+1}{2}$, an affine function of c, and $\zeta(c) = -n\lfloor t_A(c)/n \rfloor - \lfloor t_B(c)/n \rfloor$. As the cell c is an affine function of (x_j, y_i), the map $\Psi_n(x, y, z) = \big(c(x, y), (z - \lambda(c(x, y)))/n\big)$ is an affine bijection of $\mathbb{R}^3$ that maps the lifted point over c to $(c, \zeta(c))$; so $\Psi_n(\mathcal{G}_n)$ has the same f-vector as $\mathcal{G}_n$, and the same vertex cells. On the grid $|t_A(c)| \le (|u_1| + |u_2|)n + |\alpha_0| + |\alpha_1|n$, so the pair $(a, b) = (\lfloor t_A/n \rfloor, \lfloor t_B/n \rfloor)$ lies in a finite set $\mathcal{L} \subset \mathbb{Z}^2$ independent of n. The cells with a given pair are the lattice points of

$$P_{ab}(n) = \{x \in \mathbb{R}^2 :\ 0 \le x_1, x_2 \le n-1,\ na \le \mathbf{u}\cdot x + \alpha(n) \le na+n-1,\ nb \le \mathbf{w}\cdot x + \beta(n) \le nb+n-1\},$$

a polygon of the form $\{x : Mx \le ng^{ab} + h\}$, where the rows of $M \in \mathbb{Z}^{8\times 2}$ are $\pm e_1$, $\pm e_2$, $\pm\mathbf{u}$, $\pm\mathbf{w}$, the vector $h \in \mathbb{Q}^8$ does not depend on (a, b) or n, and $g^{ab} \in \mathbb{Q}^8$ does not depend on n and satisfies $\delta g^{ab} \in \mathbb{Z}^8$. Since $\operatorname{conv}(X \times \{\zeta\}) = \operatorname{conv}(X) \times \{\zeta\}$ for $X \subset \mathbb{R}^2$,

$$\Psi_n(\mathcal{G}_n) = \operatorname{conv} \bigcup_{(a,b)\in\mathcal{L}} \operatorname{conv}\big(P_{ab}(n) \cap \mathbb{Z}^2\big) \times \{-na - b\}. \tag{5}$$

Since $P^{ab}_\infty = \{x : Mx \le g^{ab}\} \subseteq [0, 1]^2$ is bounded, the following lemma applies to each $P_{ab}(n)$.

Lemma D.1 (Integer hulls of a linear family of polygons)**.** *Let $M \in \mathbb{Z}^{K\times 2}$ have rows $\mathbf{r}_k$, let $g, h \in \mathbb{Q}^K$, put $P_n = \{x \in \mathbb{R}^2 : Mx \le ng + h\}$, and suppose that $P_\infty = \{x : Mx \le g\}$ is bounded. Let D be the least common multiple of the denominators q_v of the vertices v of P_∞ (q_v is the least positive integer with $q_v v \in \mathbb{Z}^2$; $D = 1$ if $P_\infty = \emptyset$). There are n_1 and, for each $r \in \{0, \dots, D-1\}$, a finite set $Z_r \subset \mathbb{Q}^2 \times \mathbb{Q}^2$ such that for all $n \ge n_1$ with $n \equiv r \pmod D$ the points $np+q$, $(p, q) \in Z_r$, lie in $P_n \cap \mathbb{Z}^2$, and $\operatorname{conv}(P_n \cap \mathbb{Z}^2) = \operatorname{conv}\{np + q : (p, q) \in Z_r\}$.*

This is a planar case of a theorem of Calegari and Walker [8, Thm. 3.5], which says that the vertices of the integer hulls of a rational family of polyhedra of size $O(n)$ have eventually quasi-polynomial coordinates; the short direct proof below also gives the explicit period D, which yields the period δL in Proposition 6.7.

Proof. If $P_\infty = \emptyset$, Farkas' lemma gives $y \ge 0$ with $yM = 0$ and $y \cdot g < 0$; then $0 = yMx \le n\, y \cdot g + y \cdot h$ fails for large n, so $P_n = \emptyset$, and we take $Z_r = \emptyset$. Otherwise, for a vertex v of P_∞ let $I_v = \{k : \mathbf{r}_k \cdot v = g_k\}$, $C_v = \{y : \mathbf{r}_k \cdot y \le h_k\ (k \in I_v)\}$ and $T_v = \{y : \mathbf{r}_k \cdot y \le 0\ (k \in I_v)\}$.

(i) $P_n \subseteq nv + C_v$ for all n, since $\mathbf{r}_k \cdot (x - nv) = \mathbf{r}_k \cdot x - ng_k$ for $k \in I_v$.

(ii) Let $\varepsilon > 0$ be the least slack $g_k - \mathbf{r}_k \cdot v$ over all vertices v and $k \notin I_v$ ($\varepsilon = \infty$ if there is no such pair), and $\nu = \max_k |\mathbf{r}_k|$. If $y \in C_v$, $|y| \le \rho$, $n \ge 1$ and $n\varepsilon \ge \nu\rho - h_k$ for all k, then $nv + y \in P_n$: for $k \notin I_v$, $\mathbf{r}_k \cdot (nv + y) \le n(g_k - \varepsilon) + \nu\rho \le ng_k + h_k$.

(iii) If some C_v is empty, $P_n = \emptyset$ for all n by (i), and we take $Z_r = \emptyset$. Otherwise $C_v = Q_v + T_v$ with Q_v a polytope and T_v the cone spanned by integer vectors $y_1, \dots, y_s$ (Minkowski–Weyl). Put $B_v = \{\sum_l \theta_l y_l : 0 \le \theta_l \le 1\}$ and $F_v(n) = (nv + Q_v + B_v) \cap \mathbb{Z}^2$. Every lattice point $x = nv + q + \sum_l \lambda_l y_l$ of $nv + C_v$ ($q \in Q_v$, $\lambda_l \ge 0$) lies in $F_v(n) + T_v$: subtract $\sum_l \lfloor \lambda_l \rfloor y_l$.

(iv) If $n \equiv r' \pmod{q_v}$, then $(n - r')v \in \mathbb{Z}^2$, so $F_v(n) = nv + E_v(r')$ with the finite set $E_v(r') = F_v(r') - r'v \subset Q_v + B_v$.

(v) Let ρ be the largest norm of a point of the sets $Q_v + B_v$, and $n_1 \geq 1$ with $n_1\varepsilon \geq \nu\rho - h_k$ for all k. By (ii), $F_v(n) \subseteq P_n \cap \mathbb{Z}^2$ for $n \geq n_1$.

(vi) Let $n \geq n_1$, $H_n = \operatorname{conv}(P_n \cap \mathbb{Z}^2)$ and $R_n = \operatorname{conv} \bigcup_v F_v(n)$, so that $R_n \subseteq H_n$ by (v); we may assume $P_n \cap \mathbb{Z}^2 \neq \emptyset$. For a linear function ℓ choose a vertex v of P_∞ at which ℓ is maximal on P_∞. For $y \in T_v$ and small $\eta > 0$ the point $v + \eta y$ lies in P_∞, as the rows $k \notin I_v$ have positive slack at v; so $\ell \leq 0$ on T_v. By (i) and (iii), $P_n \cap \mathbb{Z}^2 \subseteq F_v(n) + T_v$, so $\max_{H_n} \ell \leq \max_{F_v(n)} \ell \leq \max_{R_n} \ell$. As this holds for every ℓ, $H_n = R_n$. Now take $Z_r = \{(v, e) : v \text{ a vertex of } P_\infty,\ e \in E_v(r \bmod q_v)\}$. □

A vertex of P_∞^{ab} solves two independent equations $\mathbf{r}_k \cdot v = g_k^{ab}$, and the determinant of two non-parallel rows of M is ± 1, $\pm u_1$, $\pm u_2$, $\pm w_1$, $\pm w_2$ or $\pm(u_1 w_2 - u_2 w_1)$; by Cramer's rule and $\delta g^{ab} \in \mathbb{Z}^8$, its denominator divides δ times the absolute value of the determinant. So the number D_{ab} of Lemma D.1 for $P_{ab}(n)$ divides δL.

Lemma D.2 (Points moving affinely)**.** *Let $\mathcal{K}$ be finite, $A_\kappa, B_\kappa \in \mathbb{Q}^3$ and $\pi_\kappa(n) = nA_\kappa + B_\kappa$ for $\kappa \in \mathcal{K}$. There is n_2 such that whether $\operatorname{conv}\{\pi_\kappa(n)\}$ is 3-dimensional is the same for all $n \geq n_2$, and, if it is, so are its f-vector and the index sets $\{\kappa : \pi_\kappa(n) \in F\}$ of its facets F.*

Proof. For $\kappa_1, \dots, \kappa_4 \in \mathcal{K}$ the determinant $\Delta_{(\kappa_1,\dots,\kappa_4)}(n)$ of the 4×4 matrix with rows $(\pi_{\kappa_l}(n), 1)$ is a polynomial in n of degree at most 3, so for $n \geq n_2$ all these determinants have constant signs; so has the relation $\pi_\kappa(n) = \pi_{\kappa'}(n)$, which holds for all n or for at most one. The hull is 3-dimensional if and only if one of these determinants is non-zero. If it is, $S \subseteq \mathcal{K}$ is the index set of a facet if and only if it contains $\kappa_1, \kappa_2, \kappa_3$ such that some $\Delta_{(\kappa_1,\kappa_2,\kappa_3,\kappa)}(n) \neq 0$, $S = \{\kappa : \Delta_{(\kappa_1,\kappa_2,\kappa_3,\kappa)}(n) = 0\}$, and the $\Delta_{(\kappa_1,\kappa_2,\kappa_3,\kappa)}(n)$ with $\kappa \notin S$ all have the same sign. A point is a vertex if and only if its index lies in at least three facet index sets; f_0 is the number of vertices up to coincidence, f_2 the number of facet index sets, and $f_1 = f_0 + f_2 - 2$. All of this depends only on the signs. □

Proof of Proposition 6.7, (a) and (b). Apply Lemma D.1 to each $P_{ab}(n)$, $(a, b) \in \mathcal{L}$, let n_1 be the largest of the thresholds and $D = \operatorname{lcm}(\delta, D_{ab} : (a, b) \in \mathcal{L})$, which divides δL. For $r \in \{0, \dots, D-1\}$ let $\mathcal{K}_r$ be the set of (a, b, p, q) with $(a, b) \in \mathcal{L}$ and $(p, q) \in Z^{ab}_{r \bmod D_{ab}}$, and $\pi_{(a,b,p,q)}(n) = (np + q, -na - b)$. By (5) and Lemma D.1, $\Psi_n(\mathcal{G}_n) = \operatorname{conv}\{\pi_\kappa(n) : \kappa \in \mathcal{K}_r\}$ for $n \in \mathcal{N}$, $n \geq n_1$, $n \equiv r \pmod D$. So $\mathcal{G}_n$ has at most $\max_r |\mathcal{K}_r|$ vertices for $n \geq n_1$, and at most n_1^2 before; this is (a). Let n_0 be the largest of n_1 and the thresholds n_2 of Lemma D.2 for the families $(\pi_\kappa)_{\kappa \in \mathcal{K}_r}$. For $n \in \mathcal{N}$ with $n \geq n_0$ also $n + D \in \mathcal{N}$, since $\delta \mid D$, and $n + D \equiv n \pmod D$; so Lemma D.2 gives (b) with D in place of δL, and repeating it $\delta L / D$ times gives (b). □

Rubin's triangle. Comparing entries in $\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}^t = \begin{pmatrix} F_{t+1} & F_t \\ F_t & F_{t-1} \end{pmatrix}$, $t \in \mathbb{Z}$, gives $F_{-t} = (-1)^{t+1} F_t$, so that $F_t \geq 1$ for every odd t, with equality only for $t = \pm 1$, and d'Ocagne's identity

$$F_r F_{s+1} - F_{r+1} F_s = (-1)^s F_{r-s} \qquad (r, s \in \mathbb{Z}). \tag{6}$$

Proof of Lemma 6.8. Write $P_t = (F_{2t}, b - F_{2t-1})$. For $0 \leq j \leq k$ put $\mathbf{n}_j = (F_{2j}, F_{2j+1})$ and $c_j = bF_{2j+1} - 1$, so that $\mathbf{n}_k = (a, b)$ and $c_k = R$, and let H be the polygon defined by $x \geq 0$, $y \geq 0$ and $\mathbf{n}_j \cdot (x, y) \leq c_j$ for $0 \leq j \leq k$; then $H \subseteq T_k$. Every lattice point of T_k lies in H: the maximum of $\mathbf{n}_j$ on T_k is attained at a vertex, $(0, 0)$, $(R/a, 0)$ or $(0, R/b)$, and since $aF_{2j+1} - bF_{2j} = F_{2k-2j} \geq 0$ by (6), it is $F_{2j+1} R/b < bF_{2j+1}$; at a lattice point $\mathbf{n}_j \cdot (x, y)$ is an integer, hence at most c_j. By (6), $c_j - \mathbf{n}_j \cdot P_t = F_{2t-1} F_{2j+1} - F_{2t} F_{2j} - 1 = F_{2(t-j)-1} - 1 \geq 0$, with equality exactly for $t \in \{j, j+1\}$. The coordinates of P_t are non-negative, as $F_{2t-1} \leq F_{2k+1} = b$ for $0 \leq t \leq k+1$; so $(0, 0)$ and the P_t lie in H. Each of these $k+3$ points lies on two non-parallel boundary lines of H: $(0, 0)$ on $x = 0$ and $y = 0$, $P_0 = (0, b-1)$ on $x = 0$ and $\mathbf{n}_0 \cdot (x, y) = c_0$, P_t for $1 \leq t \leq k$ on the lines of $\mathbf{n}_{t-1}$ and $\mathbf{n}_t$ (with $\det(\mathbf{n}_t, \mathbf{n}_{t-1}) = F_2 = 1$ by (6)), and $P_{k+1} = (F_{2k+2}, 0)$ on $y = 0$ and the line of $\mathbf{n}_k$. So they are $k+3$ vertices of H, distinct since the F_{2t} with $t \geq 1$ are distinct and positive and $b - 1 \geq 1$; as H is defined by $k+3$ inequalities, it has no others. Hence H is the convex hull of lattice points of T_k, and $\operatorname{conv}(T_k \cap \mathbb{Z}^2) = H$. □